TRACK P (PLANETARY): BLUEPRINT

SCOTT ARMSTRONG AND ZOHAR KOMARGODSKI

Abstract. A proof plan, step by step, for the three frozen statements of track P: the properly degenerate KAM theorem of Chierchia–Pinzari (specialized as frozen), the analytic planar two-planet theorem with a moving Sun, and the fixed-Sun n-planet variant. Part 0 fixes conventions, restates every frozen definition and the three frozen theorems; the remaining parts are listed in `PLAN.md`. Results of track K's blueprint are cited as [K:`label`].

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1. Conventions, definitions and the frozen statements

This part fixes notation and restates, in mathematical form, the three frozen theorems of track P (owner approval D20, 27 October 2026) together with every frozen definition they use: the 33 definitions of `KAM/Frozen/Planetary/Defs/` and the three track-K definitions `KAM.IsDiophantine`, `KAM.hamiltonianVectorField`, `KAM.symplecticForm`. The Lean text is authoritative and the statements below transcribe it, with the same quantifier order and the same order in which constants are chosen. The D22 additions (P-EXP-SAFE, P-AMD-CONFINEMENT, P-SECULAR-BOX) are not frozen and are deliberately absent; a later part is reserved for them (`PLAN.md`, Part 14).

1.1. Conventions.

- *Ambient spaces.* All coordinate spaces are $\mathbb{R}^d = (\texttt{Fin}\, d \to \mathbb{R})$ with the max norm $|v|_\infty = \max_i |v_i|$ and Lebesgue measure; $\mathrm{Ph}(d) = \mathbb{R}^d \times \mathbb{R}^d$ carries the max norm of the two components and the product Lebesgue measure `volume`. Fourier indices are $k \in \mathbb{Z}^d$. The Lean block conventions are: for $v \in \mathbb{R}^{a+b}$, the *fast* part is $v^{\mathrm{f}} = (v_i)_{i<a}$ and the *slow* part is $v^{\mathrm{s}} = (v_{a+j})_{j<b}$.
- *Phase order.* A phase point is $z = (z_1, z_2)$ with positions (angles) z_1 and momenta (actions) z_2. In the properly degenerate setting $z = (\varphi, q; I, p)$, that is $z_1 = (\varphi, q)$ and $z_2 = (I, p)$, with $\varphi \in \mathbb{R}^a$, $q \in \mathbb{R}^b$ and $I \in \mathbb{R}^a$, $p \in \mathbb{R}^b$ (`assemble`).
- *Physical distance.* On $\mathbb{R}^2$ the physical length is $|x|_2 = \sqrt{x_1^2 + x_2^2}$, never the max norm; $x \cdot y = x_1 y_1 + x_2 y_2$. Neighbourhoods in phase space and the norms $|\cdot|$ of tori data use the max norm; physical formulas use $|\cdot|_2$.
- *Bodies.* Positions of n planets are $q \in \mathbb{R}^{2n}$; planet i has position $q^{(i)} = (q_{(i,1)}, q_{(i,2)}) \in \mathbb{R}^2$, where the index of the pair $(i, k) \in \texttt{Fin}\, n \times \texttt{Fin}\, 2$ is $k + 2i$ (Lean `finProdFinEquiv`).
- *Torus parameters have period one* for every torus predicate below. For the cylinder tori of P-G (Definition 1.15) the *canonical fast angles keep period* 2π (CP10 footnote 1, p. 546), so the parameterization winds by 2π in the fast directions. Consequently the frequency vector w of a torus is the physical angular frequency divided by 2π (it changes frequency constants, not the dynamics or the measure ratio; `statement.md`, "Conventions"; owner question Q-P2). For the physical tori the parameter lattice is $\mathbb{Z}^d$ and the angles $2\pi\phi$ enter only through `circularPoint`.
- *Constants and quantifier order.* In each theorem, all data precede the small parameters; the proofs must keep the order of Theorems 1.28–1.30. In particular the thresholds r_*, C, c of P-G depend on the whole fixed family $(V, W, \dots, P, \dots)$, and μ_0, d, D (resp. ε_0, d, D) are chosen before μ (resp. ε).

1.2. Track-K definitions used by track P.

The three frozen track-P theorems import exactly these frozen track-K definitions; no definition of track I or N is used.

Definition 1.1 (Diophantine vectors)**.** For $\gamma, \tau \in \mathbb{R}$ and $\omega \in \mathbb{R}^d$, ω is (γ, τ)-*Diophantine* if $\gamma/|k|_1^\tau \le |k \cdot \omega|$ for every $k \in \mathbb{Z}^d \setminus \{0\}$, where $|k|_1 = \sum_i |k_i|$ and $k \cdot \omega = \sum_i k_i \omega_i$.
Frozen Lean declaration: `KAM.IsDiophantine`
Source: HL19 (3.1), p. 14; Pos01 (1), p. 5; same as [K:`bp:p0:diophantine`].

Definition 1.2 (Hamiltonian vector field and symplectic form)**.** For $H : \mathrm{Ph}(d) \to \mathbb{R}$ and $z = (z_1, z_2)$,

$$X_H(z) = \big(\partial_{z_2} H(z),\, -\partial_{z_1} H(z)\big),$$

that is $X_H(z)_{1,i} = DH(z)(0, e_i)$ and $X_H(z)_{2,i} = -DH(z)(e_i, 0)$, with D the Fréchet derivative (`fderiv`). For $v = (v_1, v_2)$, $w = (w_1, w_2) \in \mathrm{Ph}(d)$,

$$\Omega(v, w) = \sum_i \big(v_{2,i} w_{1,i} - v_{1,i} w_{2,i}\big).$$

With z_1 the position and z_2 the momentum, $\dot z_1 = \partial_{z_2} H$, $\dot z_2 = -\partial_{z_1} H$ are the usual Hamilton equations, and $\Omega = \sum dz_{2,i} \wedge dz_{1,i}$. The condition $\Omega(Dv, Dw) = 0$ defining a Lagrangian torus is insensitive to the sign of Ω.
Frozen Lean declaration: `KAM.hamiltonianVectorField, KAM.symplecticForm`
Source: HL19 Rem. 2.1, p. 6; same as [K:`bp:p0:hamiltonian`].

1.3. **Elementary planar and index definitions.**

Definition 1.3 (Phase space)**.** $\mathrm{Ph}(d) = \mathbb{R}^d \times \mathbb{R}^d$ (canonical coordinates *(position, momentum)*).
Frozen Lean declaration: `KAM.Planetary.Phase`

Definition 1.4 (Planar length and scalar product)**.** For $v, w \in \mathbb{R}^2$: $\mathrm{sq}(v) = \sum_k v_k^2$, $|v|_2 = \sqrt{\mathrm{sq}(v)}$ and $v \cdot w = \sum_k v_k w_k$.
Frozen Lean declaration: `KAM.Planetary.planeSq, KAM.Planetary.planeLength, KAM.Planetary.planeDot`
Source: Fej04r (28), pp. 46–47 (physical Euclidean length; independent of the ambient max norm).

Definition 1.5 (Body vectors)**.** For $q \in \mathbb{R}^{2n}$ and $i < n$, $\mathrm{body}(q, i) = q^{(i)} = (q_{(i,1)}, q_{(i,2)}) \in \mathbb{R}^2$.
Frozen Lean declaration: `KAM.Planetary.bodyVector`

Definition 1.6 (Fast and slow blocks, assembly)**.** For $v \in \mathbb{R}^{a+b}$, $v^{\mathrm{f}} \in \mathbb{R}^a$ and $v^{\mathrm{s}} \in \mathbb{R}^b$ are the first a and last b coordinates. For $I, \varphi \in \mathbb{R}^a$ and $p, q \in \mathbb{R}^b$, $\mathrm{assemble}(I, \varphi, p, q) = \big((\varphi, q), (I, p)\big) \in \mathrm{Ph}(a+b)$, with the concatenated coordinate vectors.
Frozen Lean declaration: `KAM.Planetary.fast, KAM.Planetary.slow, KAM.Planetary.assemble`
Source: CP10 (1), p. 546; canonical order $(\varphi, q; I, p)$.

Definition 1.7 (Oscillator actions)**.** For $p, q \in \mathbb{R}^b$ and $j \le b$: $R_j(p, q) = (p_j^2 + q_j^2)/2$.
Frozen Lean declaration: `KAM.Planetary.oscillatorAction`
Source: CP10 (A2") and CP11 (A_2), pp. 547 and 51.

Definition 1.8 (Fast shift)**.** For $k \in \mathbb{Z}^a$, $\mathrm{shift}(k) = \big(((2\pi k_i)_{i<a}, 0_b),\, 0\big) \in \mathrm{Ph}(a+b)$: the fast angles are translated by $2\pi k$, the slow position coordinates and all momenta are unchanged.
Frozen Lean declaration: `KAM.Planetary.fastShift`

Definition 1.9 (Fast frequency map)**.** For $H_0 : \mathbb{R}^a \to \mathbb{R}$ and $I \in \mathbb{R}^a$, $\omega_{H_0}(I)_i = DH_0(I)(e_i)$ (the Fréchet derivative is 0 where H_0 is not differentiable).
Frozen Lean declaration: `KAM.Planetary.fastFrequency`
Source: CP10 (A1), p. 546.

1.4. **Definitions for the properly degenerate theorem.**

Definition 1.10 (Cylinder and its fundamental cell)**.** For $V \subset \mathbb{R}^a$ and $r \in \mathbb{R}$,

$$\mathcal{C}(V, r) = \Big\{ z \in \mathrm{Ph}(a+b) :\ z_2^{\mathrm{f}} \in V,\ \ (z_2^{\mathrm{s}})_j^2 + (z_1^{\mathrm{s}})_j^2 < r^2\ \ \forall j \le b \Big\},$$

that is $I \in V$ and $p_j^2 + q_j^2 < r^2$ for every j: a *product of discs*, not a box. Its fundamental cell for the fast angles of period 2π is

$$\mathcal{F}(V, r) = \big\{ z \in \mathcal{C}(V, r) :\ 0 \le z_{1,i}^{\mathrm{f}} < 2\pi\ \ \forall i \le a \big\}.$$

Uses require $r > 0$.
Frozen Lean declaration: `KAM.Planetary.cylinder, KAM.Planetary.cylinderCell`
Source: CP10 p. 561, below (55) (disc region; audit P1); CP10 footnote 1 (period 2π).

Definition 1.11 (The family $H_0 + \mu P$)**.** For $H_0 : \mathbb{R}^a \to \mathbb{R}$ and $P : \mathbb{R} \times \mathrm{Ph}(a+b) \to \mathbb{R}$, $H_\mu(z) = H_0(z_2^{\mathrm{f}}) + \mu P(\mu, z)$.
Frozen Lean declaration: `KAM.Planetary.degenerateHamiltonian`
Source: CP10 (1), p. 546.

Definition 1.12 (Averaged perturbation)**.** For $\mu \in \mathbb{R}$, $I \in \mathbb{R}^a$, $p, q \in \mathbb{R}^b$,

$$P_{\mathrm{av}}(\mu, I; p, q) = (2\pi)^{-a} \int_{[0,2\pi]^a} P\big(\mu, \mathrm{assemble}(I, \varphi, p, q)\big)\, d\varphi$$

(Bochner integral; $= 0$ for a non-integrable integrand, so every use asserts integrability).
Frozen Lean declaration: `KAM.Planetary.secularAverage`
Source: CP10 (3), p. 547.

Definition 1.13 (Fast diffeomorphism)**.** For H_0 and $W \subset \mathbb{R}^a$, HasFastDiffeo(H_0, W) holds if (i) H_0 is real analytic at every point of W (`AnalyticOnNhd`); (ii) $\omega_{H_0}(W)$ is open; (iii) there is $g : \mathbb{R}^a \to \mathbb{R}^a$, of class C^1 on $\omega_{H_0}(W)$, with $g(\omega_{H_0}(I)) = I$ for $I \in W$, and $g(w) \in W$ and $\omega_{H_0}(g(w)) = w$ for $w \in \omega_{H_0}(W)$. Thus ω_{H_0} is a bijection of W onto its open image with C^1 inverse g (explicit, not a totalized inverse).
Frozen Lean declaration: `KAM.Planetary.HasFastDiffeomorphism`
Source: CP10 (A1), p. 547; neighbourhood version by audit P2.

Definition 1.14 (Secular Birkhoff form (A2"))**.** Let $V \subset \mathbb{R}^a$, $r_0, \mu_0 \in \mathbb{R}$, and functions $P_0 : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}$, $\Omega : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}^b$, $\beta : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}^{b \times b}$. HasBirkhoff$(P, V, r_0, \mu_0, P_0, \Omega, \beta)$ holds if for every $\mu \in [0, \mu_0]$ and $I \in V$:

(i) for all $p, q \in \mathbb{R}^b$ with $p_j^2 + q_j^2 < r_0^2$ for all j, $\varphi \mapsto P(\mu, \mathrm{assemble}(I, \varphi, p, q))$ is integrable on $[0, 2\pi]^a$;
(ii) for every $\epsilon > 0$ there is δ with $0 < \delta \le r_0$ such that for all p, q with $p_j^2 + q_j^2 < \delta^2$ for all j,

$$\Big| P_{\mathrm{av}}(\mu, I; p, q) - \Big(P_0(\mu, I) + \sum_j \Omega_j(\mu, I) R_j + \tfrac{1}{2} \sum_{i,j} \beta_{ij}(\mu, I) R_i R_j \Big) \Big| \le \epsilon\, |(p, q)|_\infty^4,$$

where $R_j = R_j(p, q)$ and $|(p, q)|_\infty = \max(|p|_\infty, |q|_\infty)$. (The case $p = q = 0$ is included and fixes $P_{\mathrm{av}}(\mu, I; 0, 0) = P_0(\mu, I)$.)

Note that δ may depend on (μ, I, ϵ) in this definition; uniformity must be derived from the joint analyticity hypothesis of Theorem 1.28 (see `PLAN.md`, Part 5).
Frozen Lean declaration: `KAM.Planetary.HasSecularBirkhoffForm`
Source: CP10 (A2"), p. 547 (CP11 (A_2), p. 51), with the oscillator-disc guards of audit P1.

Definition 1.15 (Cylinder torus)**.** Let $H : \mathrm{Ph}(a + b) \to \mathbb{R}$, $U \subset \mathrm{Ph}(a + b)$, $w \in \mathbb{R}^{a+b}$, $K : \mathbb{R}^{a+b} \to \mathrm{Ph}(a + b)$. IsCylTorus$(H, U, w, K)$ holds if:

(i) K is real analytic on $\mathbb{R}^{a+b}$;
(ii) $K(\theta) \in U$ and H is differentiable at $K(\theta)$, for all θ;
(iii) $K(\theta + k) = K(\theta) + \mathrm{shift}\big((k_i)_{i<a}\big)$ for all $\theta \in \mathbb{R}^{a+b}$, $k \in \mathbb{Z}^{a+b}$;
(iv) if $K(\theta') = K(\theta) + \mathrm{shift}(k)$ for some $k \in \mathbb{Z}^a$, then $\theta' - \theta \in \mathbb{Z}^{a+b}$;
(v) $DK(\theta)$ is injective for all θ;
(vi) $\Omega(DK(\theta)u, DK(\theta)v) = 0$ for all θ, u, v;
(vii) for all $\theta, t \in \mathbb{R}$, $s \mapsto K(\theta + sw)$ has derivative $X_H(K(\theta + tw))$ at $s = t$.

No Diophantine condition is part of this predicate (see Definition 1.16).
Frozen Lean declaration: `KAM.Planetary.IsCylinderTorus`
Source: CP11 Thm. 11.1, pp. 50–51 (analytic conjugacy to a linear flow, Lagrangian); CP10 Thm. 1.3, p. 548.

Definition 1.16 (Cylinder Kolmogorov set)**.** For $H : \mathrm{Ph}(a + b) \to \mathbb{R}$, $V \subset \mathbb{R}^a$, $r, \mu, C \in \mathbb{R}$ and $S \subset \mathrm{Ph}(a + b)$, IsCylKol$(H, V, r, \mu, C, S)$ holds if S is Lebesgue measurable, $S \subset \mathcal{C}(V, r)$, and for every $z \in S$ there are $w \in \mathbb{R}^{a+b}$, $K : \mathbb{R}^{a+b} \to \mathrm{Ph}(a + b)$ and θ with: $K(\theta) = z$; $K(\mathbb{R}^{a+b}) \subset S$ (a whole torus); IsCylTorus$(H, \mathcal{C}(V, r), w, K)$; there are $\gamma > 0$ and $\tau > a + b$ with w (γ, τ)-Diophantine (the constant may depend on the torus); $|w^{\mathrm{f}}|_\infty \le C$ and $|w^{\mathrm{s}}|_\infty \le C\mu$.
Frozen Lean declaration: `KAM.Planetary.IsCylinderKolmogorovSet`
Source: CP11 Thm. 11.1, pp. 50–51: $\omega_1 = O(1)$, $\omega_2 = O(\mu)$, Diophantine.

1.5. **Definitions for the physical theorems.** Throughout, $G, m_0 > 0$ and the planet weights $m_i > 0$.

Definition 1.17 (Reference circular configurations and the tube)**.** For $m, a \in \mathbb{R}^n$ and $\phi \in \mathbb{R}^n$ (phases in $\mathbb{R}/\mathbb{Z}$),

$$\mathrm{circ}(\phi) = \Big(\big(a_i(\cos 2\pi\phi_i, \sin 2\pi\phi_i)\big)_i,\ \big(m_i\sqrt{Gm_0/a_i}\,(-\sin 2\pi\phi_i, \cos 2\pi\phi_i)\big)_i \Big) \in \mathrm{Ph}(2n)$$

(circular prograde Kepler configurations at coupling 0; $a_i > 0$, $G, m_0 > 0$ guard the square root and the division). For $\eta \in \mathbb{R}$ the *circular tube* is the open η-neighbourhood in the phase max norm, $U_\eta = \{z \in \mathrm{Ph}(2n) : \exists \phi \in \mathbb{R}^n,\ |z - \mathrm{circ}(\phi)|_\infty < \eta\}$.
Frozen Lean declaration: `KAM.Planetary.circularPoint, KAM.Planetary.circularTube`
Source: Fej04r Thm. 63, p. 49 (near circular coplanar Kepler motions); ours for the explicit reference point.

Definition 1.18 (No collision, pair bounds)**.** For $q \in \mathbb{R}^{2n}$: NoColl(q) means $|q^{(i)}|_2 > 0$ for all i and $|q^{(i)} - q^{(j)}|_2 > 0$ for all $i \neq j$ (heliocentric configurations, the Sun at the origin). For N points $Q_0, \ldots, Q_{N-1} \in \mathbb{R}^2$, PairBounds$(d, D; Q)$ means $d \le |Q_j - Q_i|_2 \le D$ for all $i \neq j$.
Frozen Lean declaration: `KAM.Planetary.NoCollision, KAM.Planetary.PairBounds`
Source: Fej04r §6.1, pp. 45–47.

Definition 1.19 (Actual masses, barycentric reconstruction)**.** For $\mu \in \mathbb{R}$ let $\mathrm{mass}_\mu = (m_0, \mu m_1, \ldots, \mu m_n) \in \mathbb{R}^{n+1}$ (index 0 is the Sun). For heliocentric positions $q \in \mathbb{R}^{2n}$ and scaled momenta $p \in \mathbb{R}^{2n}$,

$$x_0(q) = -\frac{\mu}{m_0 + \mu\sum_i m_i}\sum_i m_i q^{(i)}, \qquad x_i(q) = x_0(q) + q^{(i)},$$

$$v_0(p) = -\frac{\mu}{m_0}\sum_i p^{(i)}, \qquad v_i(p) = \frac{1}{m_i}p^{(i)} \qquad (1 \le i \le n),$$

with the Sun at index 0 and planet i at index i in the order of `Fin.cons`. The denominators are positive for $m_i > 0$, $\mu > 0$.
Frozen Lean declaration: `KAM.Planetary.actualMasses, KAM.Planetary.barycentricPositions, KAM.Planetary.barycentricVelocities`
Source: Fej04r §6.1, (28), pp. 46–47; zero total momentum and barycentre; Fej07e p. 3.

Definition 1.20 (Newton solutions)**.** Let $N \in \mathbb{N}$, $M \in \mathbb{R}^N$ and $Q, V : \mathbb{R} \to (\mathbb{R}^2)^N$. IsNewton$(G, M, Q, V)$ holds if for all $t \in \mathbb{R}$ and $i \neq j$ (all indices $< N$): (i) $|Q_j(t) - Q_i(t)|_2 > 0$; (ii) Q_i has derivative $V_i(t)$ at t; (iii) V_i has derivative

$$\sum_{j \neq i} \frac{GM_j}{|Q_j(t) - Q_i(t)|_2^3}\,(Q_j(t) - Q_i(t)) \quad \text{at } t$$

(the all-time Newton equations; `HasDerivAt`).
Frozen Lean declaration: `KAM.Planetary.IsNewtonSolution`
Source: Fej04r (1), §6.1, with G restored.

Definition 1.21 (Fixed-Sun solutions)**.** For $Q, V : \mathbb{R} \to (\mathbb{R}^2)^n$ and $\varepsilon \in \mathbb{R}$, the predicate IsFixedSol$(G, m_0, m, \varepsilon, Q, V)$ holds if for all t and $i \neq j$: $|Q_i(t)|_2 > 0$, $|Q_j(t) - Q_i(t)|_2 > 0$, Q_i has derivative $V_i(t)$ and V_i has derivative

$$-\frac{Gm_0}{|Q_i(t)|_2^3}\,Q_i(t) + \varepsilon\sum_{j \neq i}\frac{Gm_j}{|Q_j(t) - Q_i(t)|_2^3}(Q_j(t) - Q_i(t)).$$

The Sun is the constant origin.
Frozen Lean declaration: `KAM.Planetary.IsFixedSolution`
Source: Ours (fixed-center law obtained by differentiating H_{fix}), after Fej04r (28).

Definition 1.22 (Moving-Sun Hamiltonian)**.** For $z = (q, p) \in \mathrm{Ph}(2n)$ and $\mu \in \mathbb{R}$,

$$H_{\mathrm{mov}}(z) = \sum_i \Big[\big(\tfrac{1}{m_i} + \tfrac{\mu}{m_0}\big) \tfrac{|p^{(i)}|_2^2}{2} - \frac{G m_0 m_i}{|q^{(i)}|_2} \Big] + \mu \sum_{i<j} \Big[\frac{p^{(i)} \cdot p^{(j)}}{m_0} - \frac{G m_i m_j}{|q^{(i)} - q^{(j)}|_2} \Big].$$

It is the reduced Hamiltonian divided by μ, with G restored, and every term is kept: the Sun moves, and neither the self kinetic term $\mu|p^{(i)}|_2^2/(2m_0)$ nor the cross term is dropped. Equivalently $H_{\mathrm{mov}} = H_{\mathrm{fix}}\big|_{\varepsilon=\mu} + \mu|\sum_i p^{(i)}|_2^2/(2m_0)$ (a calculation for a later part).

Frozen Lean declaration: `KAM.Planetary.movingHamiltonian`

Source: Fej04r (28), pp. 46–47, with Fej07e p. 3 (indices $m_j m_k$ corrected) and G restored.

Definition 1.23 (Fixed-Sun Hamiltonian)**.** For $z = (q, p) \in \mathrm{Ph}(2n)$ and $\varepsilon \in \mathbb{R}$,

$$H_{\mathrm{fix}}(z) = \sum_i \Big[\frac{|p^{(i)}|_2^2}{2m_i} - \frac{G m_0 m_i}{|q^{(i)}|_2} \Big] - \varepsilon \sum_{i<j} \frac{G m_i m_j}{|q^{(i)} - q^{(j)}|_2}.$$

The Sun is the origin; only the direct planet–planet interaction is kept, with coefficient ε.

Frozen Lean declaration: `KAM.Planetary.fixedHamiltonian`

Source: Ours (Newtonian fixed-center model; "our variant of Fej04r (28)"); `statement.md`, P-FIXED.

Definition 1.24 (Physical torus)**.** For $H : \mathrm{Ph}(d) \to \mathbb{R}$, $w \in \mathbb{R}^d$ and $K : \mathbb{R}^d \to \mathrm{Ph}(d)$, $\mathrm{IsPhysTorus}(H, w, K)$ holds if:

(i) K is real analytic on $\mathbb{R}^d$;
(ii) H is differentiable at $K(\theta)$ for all θ;
(iii) $K(\theta + k) = K(\theta)$ for $k \in \mathbb{Z}^d$;
(iv) $K(\theta) = K(\theta')$ implies $\theta' - \theta \in \mathbb{Z}^d$;
(v) $DK(\theta)$ is injective;
(vi) $\Omega(DK(\theta)u, DK(\theta)v) = 0$;
(vii) there are $\gamma > 0$ and $\tau > d - 1$ with w (γ, τ)-Diophantine;
(viii) for all θ, t, $s \mapsto K(\theta + sw)$ has derivative $X_H(K(\theta + tw))$ at $s = t$.

(For such K the image is a compact embedded analytic Lagrangian torus. Track K's primary-torus predicate has affine winding and is not reused. Note $\tau > d - 1$ here against $\tau > a + b$ in Definition 1.16.)

Frozen Lean declaration: `KAM.Planetary.IsPhysicalTorus`

Source: HL19 (2.1), (2.3), pp. 6–7 (modified to period-one embeddings); CP09 Thm. 4, p. 8; ours.

Definition 1.25 (Moving-Sun Kolmogorov set)**.** For $G, m_0 > 0$, $m, a \in \mathbb{R}^n$, $\mu, \eta, d, D \in \mathbb{R}$, $S \subset \mathrm{Ph}(2n)$: $\mathrm{MovKol}(G, m_0, m, a, \mu, \eta, d, D, S)$ holds if S is measurable, $S \subset U_\eta$, and every $z \in S$ has w, K, θ with $K(\theta) = z$, $K(\mathbb{R}^{2n}) \subset S$, $\mathrm{IsPhysTorus}(H_{\mathrm{mov}}, w, K)$ and, for every $\phi \in \mathbb{R}^{2n}$:

(a) $\mathrm{NoColl}\big(K(\phi)_1\big)$;
(b) $\mathrm{IsNewton}\big(G, \mathrm{mass}_\mu,\ t \mapsto x(K(\phi+tw)_1),\ t \mapsto v(K(\phi+tw)_2)\big)$, where x, v are the barycentric reconstructions of Definition 1.19;
(c) $\mathrm{PairBounds}\big(d, D;\ x(K(\phi)_1)\big)$ (all pairs including the Sun; as ϕ is arbitrary this covers every point $K(\phi + tw)$ of every orbit, hence all times).

Frozen Lean declaration: `KAM.Planetary.MovingKolmogorovSet`

Source: Fej04r Prop. 77 and Fej07e §7, pp. 3–4, with CP09 Thm. 4; the physical conclusions are ours (`statement.md`).

Definition 1.26 (Fixed-Sun Kolmogorov set)**.** For ε, η, d, D and $S \subset \mathrm{Ph}(2n)$ the predicate $\mathrm{FixKol}(G, m_0, m, a, \varepsilon, \eta, d, D, S)$ holds

if S is measurable, $S \subset U_\eta$, and every $z \in S$ has w, K, θ with $K(\theta) = z$, $K(\mathbb{R}^{2n}) \subset S$, IsPhysTorus$(H_{\mathrm{fix}}, w, K)$ and for every ϕ: NoColl$(K(\phi)_1)$; IsFixedSol$\big(G, m_0, m, \varepsilon,\ t \mapsto (K(\phi + tw)_1^{(i)})_i,\ t \mapsto (\frac{1}{m_i} K(\phi + tw)_2^{(i)})_i\big)$; PairBounds$\big(d, D; (0, K(\phi)_1^{(1)}, \dots, K(\phi)_1^{(n)})\big)$, with the Sun at the origin as point 0.

Frozen Lean declaration: `KAM.Planetary.FixedKolmogorovSet`

Source: Ours (new deduction; Fej04r Lemma 64 and Prop. 77, CP09 Thm. 4).

Remark 1.27 (Lean totality). The frozen definitions use Lean total functions, and the frozen theorems guard each of them as follows; proofs must establish these guarded facts before using the defining formulas.

- *Fréchet derivatives* (ω_{H_0}, X_H, DK) are 0 at points of non-differentiability. ω_{H_0} is used only through `HasFastDiffeomorphism`, which asserts analyticity of H_0 on $W \supset \overline{V}$; X_H appears only in the clauses (vii)/(viii) of the torus predicates, which also assert differentiability of H on the range of K (for the cylinder tori at points of $\mathcal{C}(V, r)$, where H_μ is analytic because $r < r_0$ and $\mu \in [0, \mu_0]$, by the hypotheses on P and H_0); DK is guarded by analyticity of K. Injectivity of DK and $\Omega(DK\cdot, DK\cdot) = 0$ are asserted for the genuine derivative.
- *Bochner integrals* (P_{av}) are 0 for non-integrable integrands; Definition 1.14 (i) asserts integrability on the disc $p_j^2 + q_j^2 < r_0^2$, and every point used in (ii) has $\delta \le r_0$.
- *Square roots and divisions*: $\sqrt{\cdot}$ in $|\cdot|_2$ has nonnegative radicand; $1/|\cdot|_2$ and $1/|\cdot|_2^3$ occur only after the collision-free clauses (a), IsNewton (i), IsFixedSol (the first two conjuncts) in the same conjunction; in circ, $\sqrt{Gm_0/a_i}$ and $1/a_i$ are guarded by $G, m_0, a_i > 0$; in x_0 and v_0 the denominators $m_0 + \mu \sum m_i$ and m_0 are positive since $\mu > 0$; the factor $1/m_i$ needs $m_i > 0$.
- *Real powers and logarithms*: $r^6/(\log(1/r))^c$ with $0 < r < r_* < 1$ has positive base $\log(1/r) > 0$; $\sqrt{r}$ has $r > 0$; $|k|_1^\tau$ in the Diophantine condition has $|k|_1 \ge 1$ for $k \ne 0$.
- *Truncated subtraction*: the lower bound is written with `ENNReal.ofReal`, so it is the real number $1 - C\sqrt{r}$ when that is positive and 0 otherwise; the strict inequality ofReal$(1 - C\sqrt{r})$ meas $\mathcal{F}$ $<$ meas$(S \cap \mathcal{F})$ is then nontrivial only when $1 - C\sqrt{r} > 0$, but must be proved for every $r < r_*$ as stated (it follows from $0 <$ meas$(S \cap \mathcal{F})$ when $1 - C\sqrt{r} \le 0$).
- *Measure*: meas is Lebesgue (outer) measure; the sets $\mathcal{F}(V, r)$ are Borel and S is measurable by the conclusion; meas $\mathcal{F}(V, r) < \infty$ since V is bounded.

1.6. **The frozen theorems.**

Theorem 1.28 (P-G: corrected properly degenerate KAM)**.** ***Data (all fixed first).*** *Integers $a, b \ge 1$. A set $V \subset \mathbb{R}^a$ that is open, connected and bounded; an open set $W \subset \mathbb{R}^a$ with $\overline{V} \subset W$; a boundary collar condition: there are $A, d > 0$ such that for all $0 < t < d$*

$$\mathrm{meas}\{I \in \mathbb{R}^a :\ \exists J \in \partial V,\ |I - J|_\infty < t\} \le A\, t.$$

Reals $r_0, \mu_0, B > 0$. Functions $H_0 : \mathbb{R}^a \to \mathbb{R}$, $P : \mathbb{R} \times \mathrm{Ph}(a + b) \to \mathbb{R}$, $P_0 : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}$, $\Omega : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}^b$, $\beta : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}^{b \times b}$.

Hypotheses.

(H1) *$(\mu, z) \mapsto P(\mu, z)$ is real analytic at every point of $[0, \mu_0] \times \overline{\mathcal{C}(V, r_0)}$ (jointly, as a function on $\mathbb{R} \times \mathrm{Ph}(a + b)$);*

(H2) *for $\mu \in [0, \mu_0]$, $z \in \mathcal{C}(V, r_0)$ and $k \in \mathbb{Z}^a$: $P(\mu, z + \mathrm{shift}(k)) = P(\mu, z)$;*

(H3) HasFastDiffeo(H_0, W) *(Definition 1.13);*

(H4) HasBirkhoff$(P, V, r_0, \mu_0, P_0, \Omega, \beta)$ *(Definition 1.14);*

(H5) *$|\det \beta(\mu, I)| \ge B$ for all $\mu \in [0, \mu_0]$, $I \in V$;*

(H6) *$\beta(\mu, I)$ is symmetric for all $\mu \in [0, \mu_0]$, $I \in V$.*

Conclusion. *There exist r_*, C, c with $0 < r_* < r_0$, $r_* < 1$, $C > 0$, $c > 0$ such that for all r, μ with*

$$0 < r < r_*, \qquad 0 < \mu < \mu_0, \qquad \mu < \frac{r^6}{(\log(1/r))^c},$$

there is $S \subset \mathrm{Ph}(a+b)$ *with* $\mathrm{IsCylKol}(H_\mu, V, r, \mu, C, S)$ *(with* $H_\mu = H_0 + \mu P(\mu, \cdot)$ *of Definition 1.11) and*

$$0 < \mathrm{meas}(S \cap \mathcal{F}(V,r)), \qquad \mathrm{ofReal}(1 - C\sqrt{r})\ \mathrm{meas}\,\mathcal{F}(V,r) < \mathrm{meas}(S \cap \mathcal{F}(V,r)) < \mathrm{meas}\,\mathcal{F}(V,r).$$

Here r_*, C, c *depend on all the data and hypotheses above and not on* r *or* μ*;* C *is simultaneously the frequency bound of Definition 1.16 and the measure constant.*

Frozen Lean declaration: `KAM.Planetary.properlyDegenerateKAM`

Source: CP11 Thm. 11.1, pp. 50–51, whose proof is CP10 Thm. 1.3, p. 548 (with $c_* = 2b$*,* $\rho = r$*). It is* not *the printed theorem: the neighbourhood* W *(CP10 p. 556, (58) p. 562), the collar (CP10 p. 561, below (56)), the product of discs (CP10 p. 561, below (55)), joint analyticity of the family (CP10 footnote 3, p. 546 asks only* C^1 *uniform dependence) and the* 2π *period are corrected-contract premises (audit P1–P4; D-PG-1 to D-PG-5).*

Source gap: CP11 Thm. 11.1 states $\mathrm{meas}\,P_\epsilon > \mathrm{meas}\,K$ *without proof; the strict inequality* $\mathrm{meas}(S \cap \mathcal{F}) < \mathrm{meas}\,\mathcal{F}$ *is obtained in the plan by deleting a small positive-measure subfamily of tori (Part 5), which is possible because the lower bound has slack. CP10 Thm. 1.3 also needs* $P_{\mathrm{av}} - N = O(\epsilon^5)$ *uniformly (CP10 (60), p. 562), which follows from (H1), (H2), (H4) and compactness but is not proved in the source.*

Theorem 1.29 (P-PLANAR: two planar planets with a moving Sun)**.** *Let* $G, m_0 > 0$*; let* $m, a : \mathbb{F}in\,2 \to \mathbb{R}$ *(planets* $0, 1$*;* m *the planet weights,* a *the axes) with* $m_i > 0$ *and* $a_i > 0$ *for* $i = 0, 1$ *and* $a_0 < a_1$ *(strictly ordered axes, no numerical separation); let* $\eta > 0$*. Then there exist* μ_0, d, D *with* $0 < \mu_0$ *and* $0 < d < D$ *such that for every* μ *with* $0 < \mu < \mu_0$ *there is* $S \subset \mathrm{Ph}(2 \cdot 2)$ *with*

$$0 < \mathrm{meas}\,S < \infty \quad \textit{and} \quad \mathrm{MovKol}(G, m_0, m, a, \mu, \eta, d, D, S) \qquad (n = 2).$$

The order is: data (G, m_0, m, a, η)*, then* (μ_0, d, D)*, then* μ*, then* S*. The tori are* 4*-dimensional in the* 8*-dimensional* $\mathrm{Ph}(4)$*.*

Frozen Lean declaration: `KAM.Planetary.planarTwoPlanetKAM`

Source: CP09 Thm. 4, p. 8 with Fej04r Prop. 77, pp. 56–58, using the recorded source-proof repairs (`statement.md`*, P5–P7); Fej07e p. 3–4 supplies the smooth counterpart. No threshold from CP11 §8.1 (audit P4).*

Source gap: The route is not signed off as proved: the analytic weak-nondegeneracy gate (CP09 Thm. 4 resting on Rüssmann's theorem, CP09 Thm. 3, whose source is not pinned), the continuation of eigenvalue branches (Fej04r Lemma 76 is false as printed) and the closeness of the tori to the circular torus (CP09 Thm. 4 says only "close to"); see `PLAN.md`*.*

Theorem 1.30 (P-FIXED: n planar planets around a fixed Sun)**.** *Let* $n \geq 2$*,* $G, m_0 > 0$*,* $m, a \in \mathbb{R}^n$ *with all* $m_i > 0$ *and* $a_i > 0$*,* $a_i < a_j$ *whenever* $i < j$*, and* $\eta > 0$*. Then there exist* ε_0, d, D *with* $0 < \varepsilon_0$ *and* $0 < d < D$ *such that for every* ε *with* $0 < \varepsilon < \varepsilon_0$ *there is* $S \subset \mathrm{Ph}(n \cdot 2)$ *with*

$$0 < \mathrm{meas}\,S < \infty \quad \textit{and} \quad \mathrm{FixKol}(G, m_0, m, a, \varepsilon, \eta, d, D, S).$$

The order is data (n, G, m_0, m, a, η)*, then* (ε_0, d, D)*, then* ε*, then* S*.*

Frozen Lean declaration: `KAM.Planetary.fixedSunPlanetaryKAM`

Source: Ours: "our proposed Newtonian fixed-center deduction, not a published fixed-Sun statement"; Fej04r Lemma 64 (p. 49–50) and Prop. 77, CP09 Thm. 4.

Source gap: The secular identification for the fixed Sun (equal secular quadratic forms and frequency map up to functions of the actions) and the general-n induction of Fej04r Prop. 77 ("par un raisonnement analogue", p. 58) are proof obligations; no secular nondegeneracy is a premise.

Remark 1.31 (Scope of this part). The three statements are the only frozen theorems of track P. P-SPATIAL is a sketch only and has no Lean statement. The D22 additions are under statement audit and are treated in the reserved last part.

2. Hamiltonian tools for tori: charts, period conversion, Diophantine conversion, Liouville

Track P produces tori in coordinates that are not the frozen Cartesian ones: CP's action–angle variables with angles of period 2π, oscillator action–angle discs, the Poincaré chart. This part supplies, once, the four tools that carry a torus found in such a chart to a frozen predicate ([P:`bpP:p0:cyltorus`], [P:`bpP:p0:phystorus`]):

- `torus-transport` (Theorem 2.9): an analytic embedding Ψ with $\Psi^*\Omega = c\,\Omega$ that is compatible with the angle lattice sends a lattice torus of $(H \circ \Psi)/c$ to a torus of H, clause by clause of the two frozen predicates;
- `period-convert` (Lemma 2.11 and Corollaries 2.12, 2.13): the passage between 2π-angles and the frozen period-one parameter is the chart $S_{2\pi}$, so the frozen frequency is $w = \tilde{\omega}/2\pi$ and the dictionary of [P:`bpP:p3:period-dictionary`] is a special case;
- `diophantine-convert` (Lemmas 2.15–2.16): CP10 (19) and CP09 (6) imply the frozen `KAM.IsDiophantine` with the exponents the two torus predicates require;
- `measure-symplectic` (Lemmas 2.18–2.20): conformally symplectic charts scale Lebesgue measure by c^d, and a union of whole tori that is a Lipschitz image is measurable with controlled measure.

The `ode` item is cited, not repeated (§2.7). Everything here is standard; the point of writing it is to fix signs, lattices and Lean totality exactly as frozen.

2.1. Conventions and external facts.

Phase space is $\mathrm{Ph}(d) = \mathbb{R}^d \times \mathbb{R}^d$, points $z = (z_1, z_2)$ (position, momentum), max norms, Lebesgue measure ([P:`bpP:p0:conventions`]). Let $M_0 = \left(\begin{smallmatrix} 0 & -\mathbf{1} \\ \mathbf{1} & 0 \end{smallmatrix}\right)$ and $J_0 = -M_0$. The frozen form ([P:`bpP:p0:hamsymp`]) is $\Omega(v, w) = \sum_i (v_{2,i} w_{1,i} - v_{1,i} w_{2,i}) = v^\top M_0 w$. For a linear form $\ell : \mathrm{Ph}(d) \to \mathbb{R}$ put $X_\ell := \big((\ell(0, e_i))_i, (-\ell(e_i, 0))_i\big)$; thus $X_H(z) = X_{DH(z)}$, and $X_H(z) = 0$ if H is not differentiable at z (Lean totality, [P:`bpP:p0:totality`]). Write $\iota_k := ((k, 0), 0) \in \mathrm{Ph}(d)$ for $k \in \mathbb{Z}^d$ (an integer translation of the positions) and, for $c > 0$, $S_c(z_1, z_2) := (c z_1, z_2)$.

(X1) Chain rule for `fderiv` and `HasDerivAt`; compositions of real analytic maps are real analytic (`AnalyticAt.comp`, `AnalyticOnNhd.comp`); sums and products of analytic maps are analytic.

(X2) Inverse function theorem: a C^1 map Ψ with $D\Psi(z_0)$ invertible is a homeomorphism from a neighbourhood of z_0 onto an open set, with C^1 inverse (`HasStrictFDerivAt.toOpenPartialHomeomorph`).

(X3) Area formula (Mathlib: `MeasureTheory.lintegral_abs_det_fderiv_eq_addHaar_image` for A measurable, Ψ differentiable and injective on A; `MeasureTheory.addHaar_image_le_lintegral_abs_det_fderiv` without injectivity): $\mathrm{meas}\,\Psi(A) = \int_A |\det D\Psi|$, resp. $\le$.

(X4) A Lebesgue measurable subset of $\mathbb{R}^N$ is the union of an F_σ set and a null set; a closed set is σ-compact; continuous images of compact sets are compact.

(X5) $\det(AB) = \det A \det B$, $\det A^\top = \det A$, and $\det M_0 = 1$.

Track K: [K:`bp:p1:holomorphic-analytic`], [K:`bp:p1:real-restriction`], [K:`bp:p1:hamiltonian-real`], [K:`bp:p1:lip-image`], [K:`bp:p1:mcshane`].

Lemma 2.1 (Algebra of Ω and X_H). *(a) $M_0^\top = -M_0$, $M_0^2 = -\mathbf{1}$, Ω is nondegenerate, and $\Omega(w, X_\ell) = \ell(w)$ for all w and all linear ℓ. Hence X_ℓ is the unique vector with this property, and $\Omega(w, X_H(z)) = DH(z)w$ for every z (both sides vanish if H is not differentiable at z). (b) If A is a real $2d \times 2d$ matrix with $\Omega(Au, Av) = c\,\Omega(u, v)$ for all u, v ($c \neq 0$), then A is invertible, $|\det A| = |c|^d$, and $X_{\ell \circ A} = c\,A^{-1} X_\ell$ for every linear form ℓ. (c) The relation $\Omega(Au, Av) = c\Omega(u, v)$ is equivalent to $A^\top(-M_0)A = c(-M_0)$: it does not depend on the sign convention for the symplectic*

form (CP10's $dI \wedge d\varphi$ is Ω; CP09 Thm. 3 prints $dx \wedge dy$ with x the angles, which is $-\Omega$; both define the same class of maps, and the vector field is the frozen X_H, not a function of the printed sign).

Proof. **Step S2.1: Matrices.** Direct from the block form; $X_\ell = J_0 \nabla \ell$ so $M_0 X_\ell = -M_0^2 \nabla \ell = \nabla \ell$, i.e. $\Omega(w, X_\ell) = w^\top M_0 X_\ell = \ell(w)$. Uniqueness: M_0 invertible.

Step S2.2: (b). $A^\top M_0 A = c M_0$ and (X5) give $(\det A)^2 = c^{2d}$. For all w: $\Omega(w, cA^{-1}X_\ell) = \Omega(Aw, AA^{-1}X_\ell) = \Omega(Aw, X_\ell) = \ell(Aw) = (\ell \circ A)(w)$, so $cA^{-1}X_\ell = X_{\ell \circ A}$ by uniqueness. (c) is the displayed matrix identity. □

Remark 2.2 (Reordering of coordinates). Part 3 writes points as (J, θ) (action, angle). The relabelling $\pi(J, \theta) = (\theta, J)$ to the frozen order (position, momentum) = (angle, action) turns $\dot\theta = \partial_J H$, $\dot J = -\partial_\theta H$ into X_H of [P:bpP:p0:hamsymp] and the form $dJ \wedge d\theta$ into Ω; no sign changes. All statements of Part 3 are read in the frozen order after applying π.

2.2. **Conformally symplectic charts.**

Definition 2.3 (Chart)**.** Let $O \subset \mathrm{Ph}(d)$ be open, $c > 0$. A *c-chart on O* is a map $\Psi : \mathrm{Ph}(d) \to \mathrm{Ph}(d)$ (total) that is real analytic at every point of O and satisfies $\Omega(D\Psi(z)u, D\Psi(z)v) = c\,\Omega(u, v)$ for $z \in O$, $u, v \in \mathrm{Ph}(d)$. A 1-chart is a *symplectic* chart. For $H : \mathrm{Ph}(d) \to \mathbb{R}$ the *pulled-back Hamiltonian* is $H^\Psi := c^{-1}\,(H \circ \Psi)$.

By Lemma 2.1(b), $D\Psi(z)$ is invertible on O, so Ψ is a local analytic diffeomorphism there ((X2); analyticity of the local inverse is not needed below).

Lemma 2.4 (Charts and Hamiltonian fields)**.** *Let Ψ be a c-chart on O, $H : \mathrm{Ph}(d) \to \mathbb{R}$, $z \in O$. Then H^Ψ is differentiable at z iff H is differentiable at $\Psi(z)$, and then*

$$D\Psi(z)\, X_{H^\Psi}(z) = X_H(\Psi(z)).$$

If H is not differentiable at $\Psi(z)$ then both fields are 0 (totality) and the identity still holds.

Proof. **Step S2.3: Differentiability.** ($\Leftarrow$) chain rule (X1). ($\Rightarrow$) by (X2) there is a C^1 local inverse Ψ_{loc}^{-1} near $\Psi(z)$ and $H = c\,H^\Psi \circ \Psi_{\mathrm{loc}}^{-1}$ near $\Psi(z)$, a composition of differentiable maps.

Step S2.4: The field. With $\ell = DH(\Psi z)$ and $A = D\Psi(z)$, $D(H^\Psi)(z) = c^{-1}\ell \circ A$, so $X_{H^\Psi}(z) = c^{-1}X_{\ell \circ A} = A^{-1}X_\ell$ by Lemma 2.1(b). Hence $AX_{H^\Psi}(z) = X_\ell = X_H(\Psi z)$. In the non-differentiable case both $D(H^\Psi)(z)$ and $DH(\Psi z)$ are 0 by the Lean convention. □

Corollary 2.5 (Orbits transfer)**.** *Let $y : J \to O$ ($J \subset \mathbb{R}$ an interval) satisfy: H^Ψ is differentiable at $y(t)$ and y has derivative $X_{H^\Psi}(y(t))$ at t, for all $t \in J$. Then $x = \Psi \circ y$ has derivative $X_H(x(t))$ at every $t \in J$ and H is differentiable at $x(t)$.*

Proof. Chain rule (X1) and Lemma 2.4. □

Remark 2.6 (What is new compared with track N). [N:bpN:p5:symplectic] and [N:bpN:p5:canonical](b),(c) give $D\Phi X_{H\circ\Phi} = X_H \circ \Phi$ and the orbit transfer for the time-one map $\Phi = X_\phi^1$ of a *holomorphic* generating function on a complex domain, with symplecticity *derived* from the flow. Here the chart is an arbitrary real analytic map (Poincaré chart, polar chart of oscillators, scalings) with symplecticity *assumed*, in the frozen real setting, with the factor c, with the frozen sign of Ω (Lemma 2.1) and with the Lean-totality guard (differentiability of H is transported, not assumed). The algebra is that of [N:bpN:p5:canonical](b); it is repeated only because the hypotheses differ. Time-one maps of Part N are charts in the sense of Definition 2.3 with $c = 1$ (Lemma 2.7).

Lemma 2.7 (Holomorphic charts, real restriction)**.** *Let $V \subset \mathbb{C}^{2d}$ be open, $\Psi : V \to \mathbb{C}^{2d}$ holomorphic, real at real points ($\Psi(V \cap \mathbb{R}^{2d}) \subset \mathbb{R}^{2d}$), and assume $D\Psi(z)^\top M_0 D\Psi(z) = cM_0$ (complex matrices) for $z \in V \cap \mathbb{R}^{2d}$. Then the restriction of Ψ to the open set $O = V \cap \mathbb{R}^{2d}$ is a c-chart on O, with real derivative $D\Psi(z)|_{\mathbb{R}^{2d}}$.*

Proof. [K:bp:p1:real-restriction] (via [K:bp:p1:holomorphic-analytic]) gives analyticity and that the real derivative is the restriction of the complex one; the identity of matrices is the defining condition of Definition 2.3 (Lemma 2.1(c)). □

2.3. **Transport of tori.**

Definition 2.8 (Lattice torus in a chart)**.** Let $O \subset \mathrm{Ph}(d)$ be open, $G : \mathrm{Ph}(d) \to \mathbb{R}$, $w \in \mathbb{R}^d$, $K : \mathbb{R}^d \to \mathrm{Ph}(d)$. $\mathrm{LT}(G, O, w, K)$ holds if

(L1) K is real analytic on $\mathbb{R}^d$, $K(\theta) \in O$ and G is differentiable at $K(\theta)$ for all θ;
(L2) $K(\theta + k) = K(\theta) + \iota_k$ for $\theta \in \mathbb{R}^d$, $k \in \mathbb{Z}^d$;
(L3) $K(\theta) = K(\theta')$ implies $\theta' - \theta \in \mathbb{Z}^d$;
(L4) $DK(\theta)$ is injective; (L5) $\Omega(DK(\theta)u, DK(\theta)v) = 0$;
(L5) $s \mapsto K(\theta + sw)$ has derivative $X_G(K(\theta + tw))$ at $s = t$, for all θ, t.

Note that (L2) is the period-one form of "angles are shifted by integers" in the chart; the frozen cylinder predicate differs from it only by the image shift in clause (iii), and the frozen physical predicate by periodicity.

Theorem 2.9 (Torus transport)**.** *Let $c > 0$, $O \subset \mathrm{Ph}(d)$ open, Ψ a c-chart on O, $H : \mathrm{Ph}(d) \to \mathbb{R}$, $G = H^\Psi = c^{-1} H \circ \Psi$, $w \in \mathbb{R}^d$, and K with $\mathrm{LT}(G, O, w, K)$. Assume the lattice compatibility*

(Q1) $O + \iota_k = O$ and $\Psi(z + \iota_k) = \Psi(z) + \sigma_k$ for $z \in O$, $k \in \mathbb{Z}^d$, for vectors $\sigma_k \in \mathrm{Ph}(d)$;
(Q2) (injectivity modulo the lattice) if $z, z' \in O$, $k \in \mathbb{Z}^d$ and $\Psi(z') = \Psi(z) + \sigma_k$, then $z' = z + \iota_m$ for some $m \in \mathbb{Z}^d$.

Put $\mathcal{K} := \Psi \circ K$. Then:

(a) (cylinder) *Let $d = a + b$ and $\sigma_k = \mathrm{shift}((k_i)_{i<a})$ (Definition 1.8). If $U \supset \Psi(K(\mathbb{R}^d))$, then* $\mathrm{IsCylTorus}(H, U, w, \mathcal{K})$ *(Definition 1.15).*
(b) (physical) *Let $\sigma_k = 0$ and assume w is (γ, τ)-Diophantine for some $\gamma > 0$, $\tau > d - 1$. Then* $\mathrm{IsPhysTorus}(H, w, \mathcal{K})$ *(Definition 1.24).*

In both cases $\mathcal{K}(\theta + tw)$ is the transport of the chart orbit, $\mathcal{K}(\mathbb{R}^d) = \Psi(K(\mathbb{R}^d))$, and $D\mathcal{K}(\theta) = D\Psi(K\theta)DK(\theta)$.

Frozen Lean declaration: `KAM.Planetary.IsCylinderTorus`, `KAM.Planetary.IsPhysicalTorus`

Source: Standard (change of canonical coordinates). Ours as a statement for the frozen predicates; no source page.

Proof. Write $\mathcal{K} = \Psi \circ K$.

Step S2.5: (i) Analyticity, range, differentiability. $K(\mathbb{R}^d) \subset O$ and Ψ is analytic on O, so $\mathcal{K}$ is analytic on $\mathbb{R}^d$ by (X1). $G = H^\Psi$ is differentiable at $K\theta$ by (L1), hence H is differentiable at $\mathcal{K}(\theta)$ by Lemma 2.4. Range: $\mathcal{K}(\theta) \in \Psi(K(\mathbb{R}^d)) \subset U$ in (a).

Step S2.6: (ii) Lattice behaviour. By (L2) and (Q1): $\mathcal{K}(\theta + k) = \Psi(K\theta + \iota_k) = \mathcal{K}(\theta) + \sigma_k$. In (a) this is clause (iii) of Def. 1.15 (with $k \in \mathbb{Z}^{a+b}$, shift by $k_f = (k_i)_{i<a}$); in (b) it is periodicity, clause (iii) of Def. 1.24.

Step S2.7: (iii) Injectivity modulo the lattice. (a): let $k \in \mathbb{Z}^a$ and $\mathcal{K}(\theta') = \mathcal{K}(\theta) + \mathrm{shift}(k)$. Then $\Psi(K\theta') = \Psi(K\theta) + \sigma_{(k,0)}$, so by (Q2) $K\theta' = K\theta + \iota_m = K(\theta + m)$ for some $m \in \mathbb{Z}^d$ (by (L2)), and (L3) gives $\theta' - (\theta + m) \in \mathbb{Z}^d$, i.e. $\theta' - \theta \in \mathbb{Z}^d$. (b): $\mathcal{K}(\theta') = \mathcal{K}(\theta)$ is the case $k = 0$ ($\sigma_0 = 0$) of the same argument.

Step S2.8: (iv) Derivative and Lagrangian property. $D\mathcal{K}(\theta) = D\Psi(K\theta)DK(\theta)$; $D\Psi(K\theta)$ is invertible (Lemma 2.1(b)), so $D\mathcal{K}(\theta)$ is injective by (L4). $\Omega(D\mathcal{K}u, D\mathcal{K}v) = c\,\Omega(DKu, DKv) = 0$ by Definition 2.3 and (L5).

Step S2.9: (v) The flow clause. Fix θ, t and let $y(s) = K(\theta + sw) \in O$. By (L6) y has derivative $X_G(y(t))$ at t and G is differentiable at $y(t)$; Corollary 2.5 (pointwise) gives that $s \mapsto \Psi(y(s)) = \mathcal{K}(\theta + sw)$ has derivative $X_H(\mathcal{K}(\theta + tw))$ at t.

Step S2.10: (vi) Diophantine clause. In (b) it is the hypothesis; in (a) none is part of IsCylTorus (see Def. 1.16 for where it is imposed; Lemmas 2.15–2.16). □

Remark 2.10 (Typical charts). (1) The polar chart of one oscillator $(\psi, R) \mapsto (q, p) = \sqrt{2R}(\sin\psi, \cos\psi)$ on $\{R > 0\}$ is a 1-chart ($dp \wedge dq = dR \wedge d\psi$, which is Ω in the frozen order), invariant under $\psi \mapsto \psi + 2\pi$; Part 5 composes it with $S_{2\pi}$ below. (2) S_c is a c-chart (Lemma 2.11). (3) Composition: if Ψ_1 is a c_1-chart on O_1, Ψ_2 a c_2-chart on $O_2 \supset \Psi_1(O_1)$, then $\Psi_2 \circ \Psi_1$ is a c_1c_2-chart on O_1 (chain rule), and $(H \circ \Psi_2)^{\Psi_1}$ with the factor c_1^{-1} followed by c_2^{-1} equals $H^{\Psi_2 \circ \Psi_1}$.

2.4. **Period conversion.** CP10 (footnote 1, p. 546: "$\mathbb{T}^n$ denotes the standard n-dimensional flat torus $\mathbb{R}^n/(2\pi\mathbb{Z}^n)$") and CP09 (Thm. 3, p. 6: "$\mathbb{T}^n$ the usual n-dimensional torus $\mathbb{R}^n/2\pi\mathbb{Z}^n$") use angles of period 2π. The frozen predicates use a parameter $\theta \in \mathbb{R}^d$ of period one. The relation is one explicit chart.

Lemma 2.11 (Period conversion)**.** *Let $c > 0$ and $S_c(z_1, z_2) = (cz_1, z_2)$.*

- *(a) S_c is a c-chart on all of* $\mathrm{Ph}(d)$ *(globally analytic, $\Omega(S_cu, S_cv) = c\,\Omega(u, v)$), $S_c(z + \iota_k) = S_c(z) + ((ck, 0), 0)$, and $|\det DS_c| = c^d$.*
- *(b)* (Hamiltonian dictionary) *For $\tilde{H} : \mathrm{Ph}(d) \to \mathbb{R}$ the pulled-back Hamiltonian is $\tilde{H}^{S_c}(z_1, z_2) = c^{-1}\tilde{H}(cz_1, z_2)$, and for every z: $S_c\, X_{\tilde{H}^{S_c}}(z) = X_{\tilde{H}}(S_cz)$. With $c = 2\pi$ and $(I, \phi) = (z_2, 2\pi z_1)$ this is $H(I, \theta) = \tilde{H}(I, 2\pi\theta)/2\pi$ of Lemma 3.5: orbits of H (angle θ) and of $\tilde{H}$ (angle $\phi = 2\pi\theta$) correspond with the* same *time; $\omega = \tilde{\omega}/2\pi$; Lagrangian for Ω iff after S_c (factor c).*
- *(c)* (Parameter conversion) *Let $\tilde{\kappa} : \mathbb{R}^d \to \mathrm{Ph}(d)$ (parameter of period 2π), $\tilde{\omega} \in \mathbb{R}^d$, $K(\theta) := \tilde{\kappa}(2\pi\theta)$, $w := \tilde{\omega}/2\pi$. Then K analytic iff $\tilde{\kappa}$ is; $DK(\theta) = 2\pi D\tilde{\kappa}(2\pi\theta)$, so injectivity of DK and the Lagrangian property (factor $4\pi^2$) hold for K iff for $\tilde{\kappa}$; $\tilde{\kappa}(\vartheta + 2\pi k) = \tilde{\kappa}(\vartheta) + \tau_k$ for all ϑ, k iff $K(\theta + k) = K(\theta) + \tau_k$; $\tilde{\kappa}(\vartheta) = \tilde{\kappa}(\vartheta') \Rightarrow \vartheta' - \vartheta \in 2\pi\mathbb{Z}^d$ iff $K(\theta) = K(\theta') \Rightarrow \theta' - \theta \in \mathbb{Z}^d$; and $s \mapsto K(\theta + sw)$ has derivative v at t iff $s \mapsto \tilde{\kappa}(2\pi\theta + s\tilde{\omega})$ has derivative v at t. In particular, for every Hamiltonian F, the frozen flow clause for (K, w) is CP's flow clause for $(\tilde{\kappa}, \tilde{\omega})$ and the frozen frequency is $w = \tilde{\omega}/2\pi$.*
- *(d)* (Frozen shift) *The frozen fast shift* $\mathrm{shift}(k_f) = ((2\pi k_f, 0_b), 0)$ *(Def. 1.8) equals $S_{2\pi}$ applied to $((k_f, 0_b), 0)$.*

Proof. **Step S2.11: (a).** $DS_c = \mathrm{diag}(c\mathbf{1}, \mathbf{1})$, so $\Omega(S_cu, S_cv) = \sum_i (u_{2,i}\, cv_{1,i} - cu_{1,i}v_{2,i}) = c\Omega(u, v)$; $\det = c^d$; the lattice formula is linearity.

Step S2.12: (b). Lemma 2.4 gives $DS_c\, X_{\tilde{H}^{S_c}} = X_{\tilde{H}} \circ S_c$. Explicitly $\partial_{z_2}\tilde{H}^{S_c} = c^{-1}\partial_2\tilde{H}$, $\partial_{z_1}\tilde{H}^{S_c} = \partial_1\tilde{H}$, which are Part 3's equations $\dot{\theta} = \partial_I H = \dot{\phi}/2\pi$, $\dot{I} = -\partial_\theta H = -\partial_\phi\tilde{H}$.

Step S2.13: (c). All claims are composition with the linear bijection $\theta \mapsto 2\pi\theta$ of $\mathbb{R}^d$ (the chain rule is not even needed for the flow clause: $K(\theta + sw) = \tilde{\kappa}(2\pi\theta + s\tilde{\omega})$ exactly). (d) is (a). □

Corollary 2.12 (Cylinder pipeline (consumed by Part 5))**.** *Let $d = a + b$, $H : \mathrm{Ph}(d) \to \mathbb{R}$, Φ a 1-chart on an open $O_0 \subset \mathrm{Ph}(d)$ with $O_0 + ((2\pi k, 0), 0) = O_0$ and $\Phi(z + ((2\pi k, 0), 0)) = \Phi(z) + \mathrm{shift}(k_f)$ for $z \in O_0$, $k \in \mathbb{Z}^d$, and $(\Phi(z') = \Phi(z) + \mathrm{shift}(k_f) \Rightarrow z' = z + ((2\pi m, 0), 0)$ for some $m \in \mathbb{Z}^d)$; put $O = S_{2\pi}^{-1}(O_0)$, $\Psi = \Phi \circ S_{2\pi}$, $G_3 = H^\Psi = (2\pi)^{-1}(H \circ \Phi)(2\pi z_1, z_2)$. If $\mathrm{LT}(G_3, O, w, K_3)$ (for instance, K_3 the torus of Prop. 3.3(a)–(d) with π of Remark 2.2 applied, and the injectivity modulo*

translations of that proposition for (L3)) and $U \supset \Psi(K_3(\mathbb{R}^d))$*, then* IsCylTorus$(H, U, w, \Psi \circ K_3)$*,* with the same w*: the frozen frequency equals Part 3's frequency* $\omega \in \mathcal{D}_{\gamma_1,\gamma_2,\tau}$*, and CP's own angular frequency is* $2\pi w$*.*

Proof. Ψ is a 2π-chart on O (Remark 2.10(3)); (Q1) holds because $\Psi(z + \iota_k) = \Phi(S_{2\pi}z + ((2\pi k, 0), 0)) = \Psi(z) + \text{shift}(k_f)$; (Q2) because $\Psi(z') = \Psi(z) + \text{shift}(k_f)$ gives $S_{2\pi}z' = S_{2\pi}z + ((2\pi m, 0), 0)$, i.e. $z' = z + \iota_m$. Apply Theorem 2.9(a). □

Corollary 2.13 (Physical pipeline (consumed by Parts 12, 13))**.** *Let* $H : \mathrm{Ph}(d) \to \mathbb{R}$*,* Φ *a* 1*-chart on an open* O_0 *with* $O_0 + ((2\pi k, 0), 0) = O_0$*,* $\Phi(z + ((2\pi k, 0), 0)) = \Phi(z)$ *and* $(\Phi(z') = \Phi(z) \Rightarrow z' = z + ((2\pi m, 0), 0)$*, some* $m \in \mathbb{Z}^d$*). Let* $\tilde\kappa : \mathbb{R}^d \to O_0$ *be a torus in the* 2π *convention of CP09 Thm. 3 for* $\tilde G = H \circ \Phi$*: i.e. the clauses (L1)–(L5) hold for* $\tilde\kappa$ *with period* 2π *as in Lemma 2.11(c) (*$\tau_k = ((2\pi k, 0), 0)$*) and* $s \mapsto \tilde\kappa(\vartheta + s\tilde\omega)$ *has derivative* $X_{\tilde G}(\tilde\kappa(\vartheta + t\tilde\omega))$ *at* t*, and* $\tilde\omega$ *is* $(\tilde\gamma, \tau)$*-Diophantine with* $\tau > d - 1$*. Then* IsPhysTorus$(H, \tilde\omega/2\pi,\ \theta \mapsto \Phi(\tilde\kappa(2\pi\theta)))$*, and* $K(\mathbb{R}^d) = \Phi(\tilde\kappa(\mathbb{R}^d))$*.*

Proof. By Lemma 2.11(c), $K_0(\theta) = \tilde\kappa(2\pi\theta)$ satisfies (L1)–(L6) for $G = \tilde G$ and $w = \tilde\omega/2\pi$ with ι_k replaced by τ_k; the argument of Theorem 2.9(b) with $\Psi = \Phi$ (a 1-chart, $\sigma = 0$, lattice τ_k in place of ι_k) applies verbatim. The frequency w is $(\tilde\gamma/2\pi, \tau)$-Diophantine (Lemma 2.15(c)). □

2.5. **Diophantine conversion.**

Definition 2.14 (Two-scale Diophantine set)**.** For $n = n_1 + n_2$, $\gamma_1, \gamma_2, \tau > 0$: $\mathcal{D}_{\gamma_1,\gamma_2,\tau}$ is the set of $\omega = (\omega_1, \omega_2) \in \mathbb{R}^{n_1} \times \mathbb{R}^{n_2}$ with $|k \cdot \omega| \geq \gamma_1 |k|_1^{-\tau}$ whenever $k_1 \neq 0$ and $|k \cdot \omega| \geq \gamma_2 |k|_1^{-\tau}$ whenever $k_1 = 0$, $k \neq 0$ ($k = (k_1, k_2) \in \mathbb{Z}^{n_1} \times \mathbb{Z}^{n_2}$). This is the set of Part 3 (§3.1); for $k_1 = 0$, $|k|_1 = |k_2|_1$.
Source: CP10 (19), p. 551 (with $\gamma_2 = \mu\bar\gamma_2$; norm $|k| = \sum|k_j|$, CP10 footnote 5, p. 547), and the definition of $\mathcal{D}_{\gamma_1,\gamma_2,\tau}$, p. 551.

Lemma 2.15 (Elementary properties of `KAM.IsDiophantine`)**.** *Let* $\omega \in \mathbb{R}^d$*,* $\gamma, \tau > 0$ *(the frozen definition is* $\gamma/|k|_1^\tau \leq |k \cdot \omega|$ *for* $k \neq 0$*, Def. 1.1).*

- *(a) (monotone)* (γ, τ)*-Diophantine implies* (γ', τ')*-Diophantine for* $0 < \gamma' \leq \gamma$*,* $\tau' \geq \tau$*. In particular* $\tau > d - 1$ *and* $\tau > a + b$ *can be arranged by raising* τ*.*
- *(b) (norms) If* $|k \cdot \omega| \geq \gamma\, \nu(k)^{-\tau}$ *for all* $k \neq 0$ *with* $\nu(k) \leq |k|_1$ *(e.g.* $\nu = |\cdot|_2$ *or* $|\cdot|_\infty$*), then* ω *is* (γ, τ)*-Diophantine.*
- *(c) (scaling) For* $s > 0$*:* ω *is* (γ, τ)*-Diophantine iff* ω/s *is* $(\gamma/s, \tau)$*-Diophantine.*
- *(d) (two scales)* $\mathcal{D}_{\gamma_1,\gamma_2,\tau} \subset \{\omega :\ \omega$ *is* $(\min\{\gamma_1, \gamma_2\}, \tau)$*-Diophantine*$\}$*; in particular, if* $\gamma_1 \geq \gamma_2$*, the set of* (γ_2, τ)*-Diophantine vectors.*

Proof. (a) For $k \neq 0$, $|k|_1 \geq 1$, so $|k|_1^\tau \leq |k|_1^{\tau'}$ (real powers of a base ≥ 1 are nondecreasing in the exponent; the Lean base is ≥ 1, so no totality issue). (b) $\nu(k)^{-\tau} \geq |k|_1^{-\tau}$. (c) $|k \cdot \omega/s| = |k \cdot \omega|/s \geq (\gamma/s)|k|_1^{-\tau}$. (d) For each $k \neq 0$ the bound in the definition is one of the two lines, each $\geq \min\{\gamma_1, \gamma_2\}|k|_1^{-\tau}$. □

Lemma 2.16 (CP10 and CP09 give the frozen Diophantine clauses)**.** *(a)* (CP10, cylinder tori) *Let* $\omega \in \mathcal{D}_{\gamma_1,\gamma_2,\tau}$ *with* $\gamma_1, \gamma_2 > 0$ *and* $\tau > n = a + b$ *(CP10:* $\tau_* > n$*, footnote 12, p. 550; Prop. 3.3 assumes* $\tau > n$*). Then* ω *is* $(\min\{\gamma_1, \gamma_2\}, \tau)$*-Diophantine, so the clause "*$\exists \gamma > 0, \tau > a + b$*,* w (γ, τ)*-Diophantine" of Def. 1.16 holds for* $w = \omega$*. If* $w = \tilde\omega/2\pi$ *with* $\tilde\omega \in \mathcal{D}_{\tilde\gamma_1,\tilde\gamma_2,\tau}$ *then* $w \in \mathcal{D}_{\tilde\gamma_1/2\pi,\tilde\gamma_2/2\pi,\tau}$*.*

(b) (CP09, physical tori) *Let* $\tilde\omega = \omega^\star(b)$ *satisfy* $|\langle k, \tilde\omega\rangle| \geq \gamma |k|_2^{-\tau}$ *for all* $k \in \mathbb{Z}^n \setminus \{0\}$ *(CP09 Thm. 3(iv), (6), p. 7:* $|k|_2$ *the Euclidean norm), with* $\gamma > 0$ *and* $\tau > n\mu \geq n$ *(*$\mu \in \mathbb{Z}_{\geq 1}$ *by CP09 Def. 2, p. 6). Then* $\tilde\omega$ *is* (γ, τ)*-Diophantine,* $w = \tilde\omega/2\pi$ *is* $(\gamma/2\pi, \tau)$*-Diophantine, and* $\tau > n - 1$*: the clause (vii) of Def. 1.24 holds with* $d = n$*.*

Proof. (a) Lemma 2.15(d) and (c). (b) Lemma 2.15(b) with $\nu = |\cdot|_2 \leq |\cdot|_1$, then (c); $\tau > n\mu \geq n > n - 1$. □

Remark 2.17 (Dimension). For the planar theorems the torus has $d = 2n$ frequencies; CP09 Thm. 3 is applied with its n equal to this d, so its $\tau > n\mu \geq d$ exceeds $d-1$ as the frozen clause requires.

2.6. **Liouville measure and unions of tori.**

Lemma 2.18 (Charts scale Lebesgue measure)**.** *Let Ψ be a c-chart on O and $A \subset O$ Lebesgue measurable. Then $\Psi(A)$ is Lebesgue measurable and* $\operatorname{meas}\Psi(A) \leq c^d \operatorname{meas} A$*, with equality if Ψ is injective on A. Symplectic charts ($c = 1$) preserve Lebesgue measure.*

Proof. **Step S2.14: Determinant.** $|\det D\Psi| = c^d$ on O (Lemma 2.1(b)).

Step S2.15: Measurability. By (X2) cover O by countably many open O_j on which Ψ is a homeomorphism onto an open set with continuous inverse Ψ_j^{-1} (Lindelöf). Write $A \cap O_j = B \cup N$, B F_σ, N null ((X4)); $\Psi(B)$ is σ-compact, $\Psi(N)$ is null because Ψ is locally Lipschitz ([K:bp:p1:lip-image] on small cubes, countable union). So $\Psi(A) = \bigcup_j \Psi(A \cap O_j)$ is measurable.

Step S2.16: Measure. (X3) gives $\operatorname{meas}\Psi(A) \leq \int_A |\det D\Psi| = c^d \operatorname{meas} A$, with equality if Ψ is injective on A. □

Lemma 2.19 (Quotient version for whole tori)**.** *Assume the hypotheses of Theorem 2.9 with $\sigma_k = 0$. Let $A \subset O$ be measurable with $A + \iota_k = A$ for all $k \in \mathbb{Z}^d$, and $Q = \{z : z_{1,i} \in [0,1)\ \forall i\}$. Then $\Psi(A)$ is measurable, $\Psi(A) = \Psi(A \cap Q)$ and $\operatorname{meas}\Psi(A) = c^d \operatorname{meas}(A \cap Q)$. Consequently $0 < \operatorname{meas}\Psi(A)$ iff $0 < \operatorname{meas} A$, and $\operatorname{meas}\Psi(A) < \infty$ if $A \cap Q$ is bounded (e.g. the momenta of A are bounded).*

Proof. $A = \bigcup_m (A \cap Q) + \iota_m$, and $\Psi((A \cap Q) + \iota_m) = \Psi(A \cap Q)$ by (Q1) with $\sigma = 0$. If $z, z' \in A \cap Q$ and $\Psi(z) = \Psi(z')$, (Q2) gives $z' = z + \iota_m$ and $z, z' \in Q$ forces $m = 0$: Ψ is injective on $A \cap Q$; Lemma 2.18 gives $\operatorname{meas}\Psi(A \cap Q) = c^d \operatorname{meas}(A \cap Q)$. Finally $\operatorname{meas} A = \sum_m \operatorname{meas}((A \cap Q) + \iota_m)$ is 0 iff $\operatorname{meas}(A \cap Q) = 0$. □

Lemma 2.20 (Unions of whole tori as Lipschitz images)**.** *Let $E \subset \mathbb{R}^d$ be Lebesgue measurable, $A = E \times [0,1)^d \subset \mathbb{R}^{2d}$, $\mathcal{F} : A \to \mathbb{R}^{2d}$ with $|\mathcal{F}(x) - \mathcal{F}(y)| \leq \Lambda |x - y|$ (max norms). Then $\mathcal{F}(A)$ is Lebesgue measurable and* $\operatorname{meas}\mathcal{F}(A) \leq \Lambda^{2d} \operatorname{meas} E$*. If moreover $|\mathcal{F}(x) - \mathcal{F}(y)| \geq \lambda |x - y|$ with $\lambda > 0$, then* $\operatorname{meas}\mathcal{F}(A) \geq \lambda^{2d} \operatorname{meas} E$*; in particular $0 < \operatorname{meas}\mathcal{F}(A)$ iff $0 < \operatorname{meas} E$, and $\operatorname{meas}\mathcal{F}(A) < \infty$ iff* $\operatorname{meas} E < \infty$*. If $\mathcal{F}(\omega, \vartheta) = \varphi_\omega(\vartheta)$ with $\varphi_\omega(\vartheta + k) = \varphi_\omega(\vartheta) + \iota_k$, then $\bigcup_{\omega \in E} \varphi_\omega(\mathbb{R}^d) = \bigcup_{k \in \mathbb{Z}^d} (\mathcal{F}(A) + \iota_k)$ is measurable.*

Proof. **Step S2.17: Extension.** McShane ([K:bp:p1:mcshane], componentwise in the max norm) gives a Λ-Lipschitz $\hat{\mathcal{F}} : \mathbb{R}^{2d} \to \mathbb{R}^{2d}$ extending $\mathcal{F}$.

Step S2.18: Measurability. A is measurable; $A = B \cup N$ with B F_σ, N null ((X4)); $\hat{\mathcal{F}}(B)$ is σ-compact and $\operatorname{meas}\hat{\mathcal{F}}(N) = 0$ by [K:bp:p1:lip-image]; so $\mathcal{F}(A) = \hat{\mathcal{F}}(B) \cup \hat{\mathcal{F}}(N)$ is measurable (completeness of Lebesgue measure).

Step S2.19: Bounds. [K:bp:p1:lip-image]: $\operatorname{meas}\mathcal{F}(A) \leq \Lambda^{2d} \operatorname{meas} A = \Lambda^{2d} \operatorname{meas} E$. Under the lower bound $\mathcal{F}$ is injective with λ^{-1}-Lipschitz inverse on $\mathcal{F}(A)$, so $\operatorname{meas} E = \operatorname{meas}\mathcal{F}^{-1}(\mathcal{F}(A)) \leq \lambda^{-2d} \operatorname{meas}\mathcal{F}(A)$ (same lemma). The last claim is the periodicity and countability of $\mathbb{Z}^d$. □

Remark 2.21 (Use). For P-PLANAR/P-FIXED the chart-level family is $X_0(b, \xi)$ of CP09 Thm. 3(ii) (bi-Lipschitz on $\mathcal{K}_\star \times \mathbb{T}^n$, $\mathcal{K}_\star$ compact): $E = \mathcal{K}_\star$, positive finite measure, and the Cartesian union is $\Psi(\cdot)$ of it, with meas = $\operatorname{meas}(A \cap Q)$ by Lemma 2.19 ($c = 1$) and $(2\pi)^d$ per period in the 2π parametrization. For P-G the family is that of Prop. 3.3, Lipschitz as stated there; the measure bounds of the cell are Part 4.

2.7. **The ODE layer (cited).** Track N's Part 3 supplies, for the three tracks, existence, uniqueness, continuation, energy and the identification of the frozen solution predicates. Track P uses only the following, always by these labels.

[N:bpN:p3:gen-uniqueness]	uniqueness on every interval for C^1 fields (orbits of a torus are *the* solutions; Lemma 2.22)
[N:bpN:p3:ham-smooth], [N:bpN:p3:ham-energy], [N:bpN:p3:ham-periodic]	X_H is C^1 where H is C^2; energy conservation; angle periodicity
[N:bpN:p3:newton-reduction], [N:bpN:p3:newton-continuation]	`IsNewtonSolution`, `IsFixedSolution` as `IsSolIn` (Part 6)
[N:bpN:p3:gen-continuation], [N:bpN:p3:gen-confinement]	only if a later part needs continuation (not used in Part 1)
[N:bpN:p5:symplectic], [N:bpN:p5:canonical]	time-one maps are symplectic; $D\Phi X_{H\circ\Phi} = X_H \circ \Phi$ (Remark 2.6)

Lemma 2.22 (Orbits on a torus)**.** *Let $H : \mathrm{Ph}(d) \to \mathbb{R}$ be C^2 on an open $W \supset \mathcal{K}(\mathbb{R}^d)$ and let $\mathcal{K}, w$ satisfy the flow clause of Defs. 1.15 (vii) or 1.24 (viii). Then $t \mapsto \mathcal{K}(\theta + tw)$ is a solution of $\dot z = X_H(z)$ in W on $\mathbb{R}$, and every solution $y : \mathbb{R} \to W$ with $y(0) = \mathcal{K}(\theta)$ equals it. $H \circ \mathcal{K}(\cdot + tw)$ is constant in t.*

Proof. The flow clause is the solution property (with values in W by hypothesis). X_H is C^1 on W ([N:bpN:p3:ham-smooth]); uniqueness is [N:bpN:p3:gen-uniqueness] on $J = \mathbb{R}$; energy is [N:bpN:p3:ham-energy]. □

Remark 2.23 (Orbit transfer to Newton). That the frozen clause (b) of MovKol and the fixed-Sun clause hold along torus orbits is Part 6: the barycentric reconstruction is a linear map composed with the orbit, the chain rule is Corollary 2.5 or elementary, and the translation of the Newton predicates is [N:bpN:p3:newton-reduction]; nothing of the existence theory is used.

2.8. **Sources, deviations, forward references.**
Sources. CP10 footnote 1 (p. 546), (19) (p. 551), footnote 5 (p. 547) and footnote 12 (p. 550), all read; CP09 Def. 2 (p. 6), Thm. 3 and (6) (pp. 6–7), read. The rest is standard (the algebra of symplectic matrices, the inverse function theorem, the area formula).
Remarks on the sources. (1) CP09 Thm. 3 prints the symplectic form as $dx \wedge dy$ with $x \in \mathbb{T}^n$ the angles; with the frozen $\Omega = \sum dz_2 \wedge dz_1$ this is $-\Omega$; the class of symplectic maps and the Lagrangian property are unaffected (Lemma 2.1(c)), and the printed Hamiltonian field in CP09 Remark 3(i) ($\xi \mapsto \xi + \omega^\star t$) is the frozen one. (2) The exponent in CP09 (6) is typeset as $|k|_2^\tau$ with a Euclidean norm; we only use $|k|_2 \le |k|_1$. (3) CP09 Thm. 3 gives no statement that $X(b, \cdot, \cdot)$ is defined on all of $\mathbb{R}^{2n}$ or injective modulo 2π translations of ξ; (L3) for $X_0(b, \cdot)$ follows from (ii) (a homeomorphism of $\mathcal{K}_\star \times \mathbb{T}^n$ onto its image).
Forward references. [P:bpP:p5:kam-application] (cylinder pipeline), Parts 6, 12, 13 (physical pipeline, orbit transfer) cite Theorem 2.9 and Corollaries 2.12, 2.13. Part 3 cites [P:bpP:p1:period-convert] (Lemma 2.11; its Lemma 3.5 is the case (b) with $c = 2\pi$). Every other cited label exists in tracks K and N.
Label index. `bpP:p1:omega-algebra`, `chart`, `chart-field`, `chart-orbit`, `hol-chart`, `lattice-torus`, `torus-transport`, `period-convert`, `cyl-pipeline`, `phys-pipeline`, `two-scale`, `dio-basic`, `dio-sources`, `liouville`, `liouville-lattice`, `torus-union`, `torus-orbit`.

3. Two-scale KAM theorem I: statement and the iterative lemma

This part (3a) states the two-scale KAM theorem of Chierchia–Pinzari (CP10 Proposition 3, pp. 554–555) in the frozen conventions and proves the iterative lemma (CP10 Lemma B.1 with Lemma B.2, pp. 565–570). Part 3b (limit, Lipschitz bounds, measure) completes the proof. *Every*

constant is recomputed here: the numerical constants differ from those of CP10 (different norms, no use of the unpinned reference [4]), the shape of $K, \hat\rho, L, \hat E$ and of the KAM condition is the same. Results of track K are cited as [K:`label`]; the weighted-norm layer and the iterative lemma of track N as [N:`label`] (*forward references*: the labels are those planned in `blueprint-N/PLAN.md`, Parts 4–6).

3.1. **Conventions, norms, constants.**

- *Period one.* $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$ with coordinates θ; Fourier characters $e^{2\pi i k\cdot\theta}$, $k \in \mathbb{Z}^n$, $|k|_1 = \sum_i |k_i|$. The symplectic form is $dJ \wedge d\theta$, the Hamilton equations are $\dot\theta = \partial_J H$, $\dot J = -\partial_\theta H$ (this is the order (action, angle) used by CP10; the frozen order $(z_1, z_2) =$ (position, momentum) is restored in Part 5). A frequency $\omega \in \mathbb{R}^n$ is a rotation number per unit time. Part 0 keeps period 2π for the canonical fast angles of P-G; the passage is the one-line dictionary of Lemma 3.5, under which *all constants of this part are invariant.*
- *Norms.* $|\cdot|$ is the max norm on $\mathbb{R}^n, \mathbb{C}^n$; $\|A\| = \max_i \sum_j |A_{ij}|$ the induced operator norm (also for rectangular matrices). Hence for a bilinear form entries and operator norm are comparable ($|A_{ij}| \le \|A\|$).
- *Complex neighbourhoods.* For $D \subset \mathbb{R}^n$ and $r > 0$: $D_r = \{I \in \mathbb{C}^n : \exists I_0 \in D,\ |I - I_0| < r\}$ and $\mathrm{Re}(D_r) = D_r \cap \mathbb{R}^n = \{x \in \mathbb{R}^n : \mathrm{dist}_\infty(x, D) < r\}$; $(D_r)_{r'} \subset D_{r+r'}$ and $D_r \subset D_{r'}$ for $r \le r'$. For $\sigma > 0$, $\mathbb{T}_\sigma = \{\theta \in \mathbb{C}^n/\mathbb{Z}^n : |\mathrm{Im}\,\theta| < \sigma/2\pi\}$ (weight σ, strip width $\sigma/2\pi$).
- *Weighted norm* (`docs/fourier-conventions.md`). For $f = \sum_k f_k(I) e^{2\pi i k\cdot\theta}$ analytic on $D_r \times \mathbb{T}_\sigma$: $\|f\|_{r,\sigma} := \sum_{k\in\mathbb{Z}^n} \sup_{I\in D_r} |f_k(I)|\, e^{\sigma|k|_1}$. It is monotone ($\|f\|_{r',\sigma'} \le \|f\|_{r,\sigma}$ for $r' \le r$, $\sigma' \le \sigma$) and $\sup_{D_r\times\mathbb{T}_\sigma} |f| \le \|f\|_{r,\sigma}$, $|f_0| \le \|f\|_{r,\sigma}$. This is the "sup–Fourier norm" of CP10 p. 551 with $\mathbb{T}^n$ of period 2π replaced by period 1 (same weights).
- *Two-scale Diophantine set.* For $n = n_1 + n_2$, $k = (k_1, k_2) \in \mathbb{Z}^{n_1} \times \mathbb{Z}^{n_2}$, $\gamma_1 \ge \gamma_2 > 0$, $\tau > 0$: $\mathcal{D}_{\gamma_1,\gamma_2,\tau} = \{\omega \in \mathbb{R}^n :\ |k\cdot\omega| \ge \gamma_1 |k|_1^{-\tau}$ if $k_1 \ne 0$, $\ |k\cdot\omega| \ge \gamma_2 |k|_1^{-\tau}$ if $k_1 = 0, k \ne 0\}$. For $\gamma_1 = \gamma_2$ this is the set of Definition 1.1; $\mathcal{D}_{\gamma_1,\gamma_2,\tau} \subset \mathcal{D}_{\gamma_2,\gamma_2,\tau}$. (CP10 (19) with $\bar\gamma_2\mu = \gamma_2$; for $k_1 = 0$ CP10 writes $|k_2|^\tau$, the same as $|k|_1^\tau$.) For $D \subset \mathbb{R}^n$ and $\omega_0 : D_\rho \to \mathbb{C}^n$ put $D^{\mathrm{dio}} := \omega_0^{-1}(\mathcal{D}_{\gamma_1,\gamma_2,\tau}) \cap D$ (this is $\mathcal{D}_{\gamma_1,\gamma_2,\tau_*}$ of CP10, "the ω_0-preimage in D").
- *Blocks.* $U = \partial^2 h$ is the $n \times n$ Hessian; $\hat U$ is the $n_2 \times n$ matrix of its last n_2 rows (the derivatives of $\omega_2 = \partial_{I_2} h$ with respect to all actions); T_1, T_2 are the first n_1 and last n_2 rows of U^{-1}.

Remark 3.1 (Misprint in the definition of $\hat U$). CP10 p. 554 defines $\hat U$ as "the $n \times n_1$ submatrix of U, i.e. the matrix with entries $\hat U_{ij} = U_{ij}$, for $n_1 + 1 \le i \le n$, $1 \le j \le n$". The size and the index ranges disagree. Only the reading above ($n_2 \times n$, rows $n_1 + 1, \dots, n$, all columns) is used in the proof: $\hat M$ bounds the variation of ω_2 (CP10 (82), via the choice of $\hat\rho$). We adopt it.

Definition 3.2 (Constants of the two-scale theorem)**.** Let $n = n_1 + n_2$, $\tau > 0$, $\gamma_1 \ge \gamma_2 > 0$, $0 < s$, $\rho > 0$, and $E, M, \hat M, \bar M > 0$. Put

$$L := \max\{\bar M,\ M^{-1},\ \hat M^{-1}\}, \qquad x := \frac{EM^2L}{\gamma_1^2}, \qquad K := \frac{6}{s}\log_+\!\big(x^{-1}\big), \quad \log_+ a := \max\{1, \log a\},$$

$$\hat\rho := \min\Big\{\frac{\gamma_1}{3MK^{\tau+1}},\ \frac{\gamma_2}{3\hat MK^{\tau+1}},\ \rho\Big\}, \qquad \hat E := \frac{EL}{\hat\rho^{\,2}}, \qquad \hat c := \max\Big\{2^{13}n,\ \frac{24^{\tau+1}}{3}\Big\}.$$

The *KAM condition* is $\hat c\hat E < 1$. (CP10 (31): $\hat c = \max\{2^8 n, 24^{\tau+1}/6\}$; the larger constants come from the max-norm Cauchy estimates and the Lipschitz bookkeeping of Parts 3a–3b. Only $\hat c = \hat c(n, \tau)$, an absolute function of n and τ, matters downstream.) *Totality.* $E > 0$ guards $\log(1/x)$, and $\log_+ \ge 1$ gives $Ks \ge 6$; $K, \hat\rho > 0$.

Source: CP10 Prop. 3, p. 555, equations (31), (32), with the changes stated.

3.2. **The theorem.**

Proposition 3.3 (Two-scale KAM theorem)**.** ***Data.*** *Integers $n_1, n_2 \geq 1$, $n = n_1 + n_2$; $\tau > n$; $\gamma_1 \geq \gamma_2 > 0$; $s, \bar{s}$ with $0 < 4s \leq \bar{s} < 1$; $\rho > 0$; an arbitrary set $D \subset \mathbb{R}^n$; $A := D_\rho$. A function $H(J,\theta) = h(J) + f(J,\theta)$,* real analytic *on $A \times \mathbb{T}_{\bar{s}+s}$, 1-periodic in θ (real on $\mathbb{R}^n \times \mathbb{R}^n$). Let $\omega_0 := \partial h$; assume ω_0 is an injective holomorphic map $A \to \mathbb{C}^n$ with $U = \partial^2 h$ nonsingular on A (hence a diffeomorphism of A onto an open set). Constants $M, \hat{M}, \bar{M}, \bar{M}_1, \bar{M}_2, E > 0$ with*

$$M \geq \sup_A \|U\|, \quad \hat{M} \geq \sup_A \|\hat{U}\|, \quad \bar{M} \geq \sup_A \|U^{-1}\|, \quad \bar{M}_i \geq \sup_A \|T_i\| \ (i = 1, 2), \quad E \geq \|f\|_{\rho, \bar{s}+s},$$

and (normalizations, always satisfiable by lowering the constants; see Remark 3.4) $\hat{M} \leq M$, $\bar{M}_i \leq \bar{M}$. Let $L, K, \hat{\rho}, \hat{E}, \hat{c}$ be as in Definition 3.2 and assume the ***KAM condition*** *$\hat{c}\hat{E} < 1$.*

Conclusions. *Put $r := 32\,\hat{E}\hat{\rho}$, $\Omega_* := \omega_0(D) \cap \mathcal{D}_{\gamma_1,\gamma_2,\tau}$ and $D^{\mathrm{dio}} := \omega_0^{-1}(\Omega_*) \cap D$. There are maps φ_ω, $\omega \in \Omega_*$,*

$$\varphi_\omega : \mathbb{R}^n \to \mathrm{Re}(D_r) \times \mathbb{R}^n, \quad \varphi_\omega(\vartheta) = \big(v(\vartheta;\omega),\ \vartheta + u(\vartheta;\omega)\big),$$

with v, u 1-periodic in ϑ, such that for every $\omega \in \Omega_*$:

- *(a) (analyticity) φ_ω is real analytic and extends holomorphically to $\{|\mathrm{Im}\,\vartheta| < \bar{s}/2\pi\}$, with $|u| \leq \frac{3}{4\pi}\hat{E}s$ there;*
- *(b) (conjugacy) $\frac{d}{dt}\varphi_\omega(\vartheta + t\omega) = X_H(\varphi_\omega(\vartheta + t\omega))$ for all $\vartheta \in \mathbb{R}^n, t \in \mathbb{R}$, where $X_H = (-\partial_\theta H, \partial_J H)$ (Hamilton equations for (J,θ));*
- *(c) (embedded Lagrangian torus) $D_\vartheta \varphi_\omega$ is injective (indeed $\|D_\vartheta u\| \leq \frac{3}{4} n\hat{E}$) and $\sum_i \big(\partial_{\vartheta_a} v_i\, \partial_{\vartheta_b}(\vartheta_i + u_i) - \partial_{\vartheta_b} v_i\, \partial_{\vartheta_a}(\vartheta_i + u_i)\big) = 0$ for all a, b;*
- *(d) (estimates) with $I^0(\omega) := \omega_0^{-1}(\omega) \in D^{\mathrm{dio}}$ and $v = (v_1, v_2) \in \mathbb{R}^{n_1} \times \mathbb{R}^{n_2}$,*

$$|v_1 - I_1^0(\omega)| \leq 20\Big(\frac{\bar{M}_1}{\bar{M}} + \frac{\hat{M}}{M}\Big)\hat{E}\hat{\rho}, \quad |v_2 - I_2^0(\omega)| \leq 20\Big(\frac{\bar{M}_2}{\bar{M}} + 1\Big)\hat{E}\hat{\rho}, \quad |u| \leq \frac{3}{4\pi}\hat{E}s \quad (\text{on } \mathbb{R}^n).$$

Moreover the map $(\vartheta, \omega) \mapsto \varphi_\omega(\vartheta)$, $\mathbb{R}^n \times \Omega_* \to \mathbb{R}^{2n}$, is injective modulo the translations $\vartheta \mapsto \vartheta + k$, $k \in \mathbb{Z}^n$ (precisely: $\varphi_\omega(\vartheta) - \varphi_{\omega'}(\vartheta') \in \{0\} \times \mathbb{Z}^n$ implies $\omega = \omega'$ and $\vartheta - \vartheta' \in \mathbb{Z}^n$), and is Lipschitz in the following sense: it factors as $\Phi \circ (\check{\imath} \circ \omega_0^{-1}, \mathrm{id})$ with $\check{\imath}$ of Lipschitz constant $\leq 1 + 2560n\hat{E}$ and Φ bi-Lipschitz in the rescaled norm of §4.1; ω_0^{-1} is $\bar{M}$-Lipschitz on Ω_* for pairs at distance $< \rho/\bar{M}$ (globally $\bar{M}\max\{1, \mathrm{diam}\, D/\rho\}$-Lipschitz if D is bounded). Finally, with $K_\Omega := \bigcup_{\omega \in \Omega_*} \varphi_\omega(\mathbb{R}^n)$ (a union of tori), $c_n := 1 + (1 + 2^n)(1 + (4/3)^n)$ and meas *the Lebesgue measure on $\mathbb{R}^n \times \mathbb{T}^n$,*

$$\mathrm{meas}\big((\mathrm{Re}(D_r) \times \mathbb{T}^n) \setminus K_\Omega\big) \leq c_n\Big(\mathrm{meas}\big((D \setminus D^{\mathrm{dio}}) \times \mathbb{T}^n\big) + \mathrm{meas}\big((\mathrm{Re}(D_r) \setminus D) \times \mathbb{T}^n\big)\Big). \tag{3.1}$$

Constant dependence. *$\hat{c}, c_n$ depend only on n, τ; $K, \hat{\rho}, \hat{E}, L$ depend only on $(n_1, n_2, \tau, \gamma_1, \gamma_2, s, \rho, E, M, \hat{M}, \bar{M})$, in the order of Definition 3.2. The sets D, h, f enter only through those constants and the hypotheses; $\bar{s}$ enters only through $4s \leq \bar{s} < 1$ and through the width of analyticity in (a).*

Not frozen itself: Part 5 consumes this statement (§4.6) to prove the frozen theorem [P:bpP:p0:pg].

Source: CP10 Prop. 3, pp. 554–555, proved in Appendix B, pp. 565–573. Constants recomputed; (35) of CP10 is corrected in (d).

Source gap: Prop. 3 has no other source ("not available in literature", CP10 p. 550); its printed proof relies on the unpinned [4, Lemma A.5] through Prop. 1/Lemma A.1 (see Lemma 3.8) and on "the inductive theorem as in Arnold [3, p. 144]" without proof (CP10 Remark 3, p. 567). The full list of misprints and repairs is in §4.7.

Remark 3.4 (Normalizations $\hat{M} \leq M$, $\bar{M}_i \leq \bar{M}$). $\hat{U}$ and T_i are sub-blocks of U, U^{-1}, so $\|\hat{U}\| \leq \|U\|$ and $\|T_i\| \leq \|U^{-1}\|$ on A: the hypotheses may be stated with $\hat{M} \leq M$ and $\bar{M}_i \leq \bar{M}$ *if one replaces* $\hat{M}$, $\bar{M}_i$ *by* $\min\{\hat{M}, M\}$, $\min\{\bar{M}_i, \bar{M}\}$; this only decreases $\hat{M}$ (which can only increase $\hat{\rho}$, hence is not harmless for $\hat{E}$: one should lower after computing $\hat{E}$). We state the proposition with the

normalization as a hypothesis, as CP10 uses it implicitly in (74), (77) and (35) ("$\le 5n\hat{E}\hat{\rho}$" needs $\bar{M}_i \le \bar{M}$). In P-G, $\hat{M} = C\mu \le M = C$ and $\bar{M}_i \le \bar{M} = C\mu^{-1}$.

Lemma 3.5 (Period dictionary)**.** *Let $\tilde{H}(I,\phi) = \tilde{h}(I) + \tilde{f}(I,\phi)$ be analytic on $A \times \{|\mathrm{Im}\,\phi| < \sigma\}$, $\phi \in (\mathbb{R}/2\pi\mathbb{Z})^n$ (CP10 convention: form $dI \wedge d\phi$, characters $e^{ik\cdot\phi}$, weighted norm $\sum_k \sup|\tilde{f}_k|e^{\sigma|k|_1}$, $\tilde{\omega} = \partial\tilde{h}$, constants $\tilde{M}, \hat{\tilde{M}}, \bar{\tilde{M}}, \tilde{E}, \tilde{\gamma}_i$, $\tilde{L}, \tilde{K}, \tilde{\rho}, \hat{\tilde{E}}$ computed by Definition 3.2 with $\tilde{\gamma}_i$ the Diophantine constants of $\tilde{\omega} = 2\pi\omega$). Put $H(I,\theta) := \tilde{H}(I, 2\pi\theta)/2\pi$. Then $h = \tilde{h}/2\pi$, $f = \tilde{f}(\cdot, 2\pi\theta)/2\pi$, $\omega = \tilde{\omega}/2\pi$, $\|f\|_{\rho,\sigma} = \|\tilde{f}\|_{\rho,\sigma}/2\pi$, and with $M = \tilde{M}/2\pi$, $\hat{M} = \hat{\tilde{M}}/2\pi$, $\bar{M} = 2\pi\bar{\tilde{M}}$, $\bar{M}_i = 2\pi\bar{\tilde{M}}_i$, $E = \tilde{E}/2\pi$, $\gamma_i = \tilde{\gamma}_i/2\pi$ one has $L = 2\pi\tilde{L}$ and $K = \tilde{K}$, $\hat{\rho} = \hat{\tilde{\rho}}$, $\hat{E} = \hat{\tilde{E}}$, $\hat{c} = \hat{\tilde{c}}$ identically. A map φ is an orbit-conjugacy for H with frequency ω iff $\phi \mapsto \varphi(\phi/2\pi)$ is one for $\tilde{H}$ with $\tilde{\omega} = 2\pi\omega$; Lagrangian for $dJ \wedge d\theta$ iff for $dI \wedge d\phi = 2\pi\, dJ \wedge d\theta$. The estimates of Proposition 3.3(d) hold verbatim for $\tilde{u} = 2\pi u$ in the form $|\tilde{u}| \le \frac{3}{2}\hat{E}s$.*

Proof. **Step S3.1: Substitution.** $\tilde{f}(I,\phi) = \sum_k \tilde{f}_k(I)e^{ik\cdot\phi}$ and $\phi = 2\pi\theta$ give $f(I,\theta) = \frac{1}{2\pi}\sum_k \tilde{f}_k(I)e^{2\pi ik\cdot\theta}$: same weights, coefficients divided by 2π. ∂_I commutes, so $U = \tilde{U}/2\pi$, $U^{-1} = 2\pi\tilde{U}^{-1}$, sub-blocks likewise.

Step S3.2: Invariants. $EM^2L/\gamma_1^2 = \frac{\tilde{E}}{2\pi}\frac{\tilde{M}^2}{4\pi^2}2\pi\tilde{L}\frac{4\pi^2}{\tilde{\gamma}_1^2} = \tilde{E}\tilde{M}^2\tilde{L}/\tilde{\gamma}_1^2$, hence $K = \tilde{K}$; $\gamma_1/(3MK^{\tau+1}) = \tilde{\gamma}_1/(3\tilde{M}K^{\tau+1})$ (same for $\gamma_2, \hat{M}$), so $\hat{\rho} = \hat{\tilde{\rho}}$; $EL/\hat{\rho}^2 = \tilde{E}\tilde{L}/\hat{\rho}^2$.

Step S3.3: Dynamics. $\dot{\phi} = \partial_I\tilde{H} = 2\pi\partial_I H = 2\pi\dot{\theta}$ and $\dot{I} = -\partial_\phi\tilde{H} = -\partial_\theta H$. (This is the `period-convert` of Part 1, [P:`bpP:p1:period-convert`], in the form needed here.) □

3.3. **External facts and imported tools.**

- (X1) *Holomorphic inverse function theorem.* An injective holomorphic $F : \Omega \to \mathbb{C}^n$ (Ω open) with $\det DF \neq 0$ is open, and $F^{-1} : F(\Omega) \to \Omega$ is holomorphic with $D(F^{-1}) = (DF)^{-1} \circ F^{-1}$.
- (X2) *Vitali–Montel.* A uniformly bounded sequence of holomorphic functions on a connected open $V \subset \mathbb{C}^n$ that converges at every point of a nonempty open subset of $V \cap \mathbb{R}^n$ (real totally real set) converges locally uniformly on V, with all derivatives.
- (X3) *Weierstrass.* Locally uniform limits of holomorphic functions are holomorphic and derivatives converge locally uniformly ([K:`bp:p1:uniform-limits`]).
- (X4) Banach fixed point ([K:`bp:p1:ext-banach`]); mean value inequality on convex sets ([K:`bp:p1:ext-mvt`]); Cauchy estimates on polydiscs ([K:`bp:p1:cauchy-open`]).
- (X5) Neumann series: $\|B\| \le \frac{1}{2}$ implies $1 + B$ invertible with $\|(1+B)^{-1}\| \le 2$.
- (X6) Lipschitz measure lemmas [K:`bp:p1:lip-image`], McShane extension [K:`bp:p1:mcshane`] (max norm).

3.4. **Two elementary lemmas.**

Lemma 3.6 (Balls in the image of a diffeomorphism)**.** *Let $\Omega \subset \mathbb{C}^n$ be open, $F : \Omega \to \mathbb{C}^n$ injective holomorphic with $\|(DF)^{-1}\| \le m$ on Ω, and $B(x_0,\delta) \subset \Omega$ (max-norm ball). Then $F(B(x_0,\delta)) \supset B(F(x_0), \delta/m)$. Moreover, if $w_1, w_2 \in B(F(x_0),\delta/m)$ then $|F^{-1}(w_1) - F^{-1}(w_2)| \le m|w_1 - w_2|$, with F^{-1} evaluated in $B(x_0,\delta)$.*

Proof. **Step S3.4: Inverse.** By (X1), $G := F^{-1} : F(\Omega) \to \Omega$ is holomorphic with $\|DG\| \le m$ and $F(B(x_0,\delta))$ is open.

Step S3.5: Path lifting. Let $w \in B(F(x_0),\delta/m)$, $\gamma(t) = F(x_0) + t(w - F(x_0))$, $\delta' = m|w - F(x_0)| < \delta$ and $T = \{t \in [0,1] : \gamma([0,t]) \subset F(B(x_0,\delta))\}$. $0 \in T$ and T is relatively open. For $t \in T$ the path $G \circ \gamma$ on $[0,t]$ is C^1 with $|(G\circ\gamma)'| \le m|w - F(x_0)|$, so $|G(\gamma(t)) - x_0| \le t\delta'$ ((X4), mean value inequality along the path).

Step S3.6: Closedness. If $t_k \uparrow t^*$, $t_k \in T$, then $G(\gamma(t_k))$ is Cauchy (Lipschitz in t), with limit x^*, $|x^* - x_0| \le \delta' < \delta$, $F(x^*) = \gamma(t^*)$ by continuity of F, so $t^* \in T$. Hence $T = [0,1]$ and $w = F(G(w)) \in F(B(x_0,\delta))$.

Step S3.7: Lipschitz bound. The segment $[w_1, w_2]$ lies in the convex ball $B(F(x_0), \delta/m) \subset F(B(x_0,\delta))$, so the mean value inequality for G gives the second claim. □

Lemma 3.7 (Lipschitz constants from sup bounds, non-convex sets)**.** *Let $S \subset \mathbb{R}^N$ be arbitrary, $r > 0$, $S_r \subset \mathbb{C}^N$ its complex r-neighbourhood in the max norm, and $\Delta : S_r \to \mathbb{C}^m$ holomorphic with $\sup_{S_r} |\Delta| \le A$. Then $|\Delta(x) - \Delta(y)| \le \frac{4NA}{r}|x - y|$ for all $x, y \in S$.*

Proof. **Step S3.8: Close pairs.** If $|x - y| < r/2$, every point z of the real segment $[x, y]$ has $|z - x| < r/2$, so the closed polydisc $\bar{P}(z, r/2) \subset S_r$ (z is within $r/2$ of $x \in S$, hence $\bar{P}(z, r/2) \subset S_r$ up to the open/closed boundary; use radius $r/2 - \epsilon$ and let $\epsilon \to 0$). By [K:bp:p1:cauchy-open], $|\partial_\ell \Delta_i(z)| \le 2A/r$, so $\|D\Delta(z)\| \le 2NA/r$, and the mean value inequality on the segment gives $|\Delta(x) - \Delta(y)| \le (2NA/r)|x - y|$.

Step S3.9: Distant pairs. If $|x - y| \ge r/2$, then $|\Delta(x) - \Delta(y)| \le 2A \le (4A/r)|x - y|$. Since $N \ge 1$, both are $\le 4NA/r$. □

3.5. Averaging with two frequency scales (the case $m = 0$, $\Lambda = \{0\}$ of CP10 Proposition 1).

Lemma 3.8 (One averaging step with two scales)**.** *Let $0 < \alpha_2 \le \alpha_1$, $D \subset \mathbb{R}^n$, $r > 0$, $\bar{s}, s > 0$, $K > 0$ with $Ks \ge 6$. Let h be analytic on D_r, $\omega = \partial h$, f analytic on $D_r \times \mathbb{T}_{\bar{s}+s}$, real analytic, 1-periodic in θ, and $E := \|f\|_{r,\bar{s}+s}$. Assume*

- *(N1) (non-resonance on the complex neighbourhood) for all $I \in D_r$ and $k \in \mathbb{Z}^n$ with $0 < |k|_1 \le K$: $|k \cdot \omega(I)| \ge \alpha_1$ if $k_1 \ne 0$ and $\ge \alpha_2$ if $k_1 = 0$;*
- *(N2) (smallness) $64e\,KE < \alpha_2 r$.*

Then there is a map $\Psi : D_{r/2} \times \mathbb{T}_{\bar{s}+s/6} \to D_r \times \mathbb{T}_{\bar{s}+s}$ ($\Psi(I', \theta') = (I, \theta)$), holomorphic, real on real points, with $\Psi - \mathrm{id}$ $\mathbb{Z}^n$-periodic in θ', symplectic for $dI \wedge d\theta$ (that is $\Psi^*(dI \wedge d\theta) = dI' \wedge d\theta'$), such that $H \circ \Psi = h + g + f_*$ on $D_{r/2} \times \mathbb{T}_{\bar{s}+s/6}$, with $g = g(I')$ independent of θ' and

$$\sup_{D_{r/2}} |g - f_0| \le E/4, \qquad \|f_*\|_{r/2,\bar{s}+s/6} \le e^{-Ks/6}E,$$

$$\alpha_1 s|I_1 - I_1'| \le 9E, \quad \alpha_2 s|I_2 - I_2'| \le 9E, \quad 2\pi\alpha_2 r|\theta - \theta'| \le 9E \qquad \text{on } D_{r/2} \times \mathbb{T}_{\bar{s}+s/6}.$$

Here $f_0(I) = \int_{\mathbb{T}^n} f(I, \theta)d\theta$ and $I = (I_1, I_2) \in \mathbb{R}^{n_1} \times \mathbb{R}^{n_2}$.

Source: CP10 Prop. 1 (p. 552) with $m = 0$, $c_0 = e/2$ so that $2^7 c_0 = 64e$, $\Lambda = \{0\}$, $d = rs$, (24)–(26); Lemma A.1 (p. 564), proof pp. 564–565.

Source gap: CP10 proves Prop. 1 only by reference to [4, Lemma A.5] (not pinned), plus the splitting (70)–(71) of the generating function for the I_1-bound. [P:bpP:p2:averaging] *is planned to derive the one-scale case $\alpha_1 = \alpha_2$ from* [N:bpN:p6:iterative]*; the two-scale case needed here is not in the plan of Part 2. This lemma is therefore an* ***interface to be supplied*** *by Part 2 (or by a variant of* [N:bpN:p6:iterative] *with two non-resonance constants).*

Plan of the proof (to be carried out in Part 2 / track N). **Step S3.10: Scaling remark.** The CP10 statements are for angles of period 2π. For $\theta = \phi/2\pi$ the small divisor $ik \cdot \omega$ becomes $2\pi ik \cdot \omega$ and ∂_θ carries a factor 2π; in every term of CP10's Poisson-bracket and generating-function estimates the two factors 2π cancel or improve (generating function $\chi_k = f_k/(2\pi ik \cdot \omega)$, $\|\chi\| \le E/(2\pi\alpha)$, then $\|\partial_\theta \chi\| \le \frac{2\pi}{e\delta'}\|\chi\|$). Hence the CP10 constants stay valid. By Lemma 3.5 one may equally apply CP10's statement to $\tilde{H} = 2\pi H(I, \phi/2\pi)$ with $\tilde{\omega} = 2\pi\omega$, $\tilde{\alpha}_i = 2\pi\alpha_i$, $\tilde{E} = 2\pi E$ (the conditions (N2) and the bounds are invariant up to the stated factors 2π in the last bound).

Step S3.11: Homological equation. Solve $\omega\cdot\partial_\theta\chi = T_K f - f_0$ by $\chi_k = f_k/(2\pi i k\cdot\omega)$, $0<|k|_1\le K$ ([N:bpN:p6:homological], [K:bp:p1:cohomological]); the splitting $k_1\neq 0/k_1=0$ gives $\|\chi^{(1)}\|\le E/(2\pi\alpha_1)$, $\|\chi^{(2)}\|\le E/(2\pi\alpha_2)$, where $\chi^{(2)}$ does not depend on θ_1 (CP10 (70)–(71)).

Step S3.12: Time-one map and remainder. Ψ is the time-one map of X_χ: domain, symplectic, estimates by [N:bpN:p5:flow-domain], [N:bpN:p5:composition], [N:bpN:p4:bracket]; the displacement bounds from the generating equations and Cauchy estimates [N:bpN:p4:cauchy]; f_* is given by the Taylor formula with integral remainder [N:bpN:p5:taylor] and $\|f-T_Kf\|\le e^{-Ks}E$ [N:bpN:p4:projections] (the weight loss $s\to s/6$ gives $e^{-Ks/6}$).

Step S3.13: Bounding g. $g-f_0$ is quadratic in E: $\le \frac{12}{11}\frac{2^7c_0KE^2}{\alpha_2 r}\le E/4$ by (N2) (CP10 (25)). □

3.6. One step of the scheme.

Lemma 3.9 (One KAM step)**.** *Let the data of Proposition 3.3 be given with $H=h+f$ on $D_\rho\times\mathbb{T}_{\bar s+s}$, constants $M,\hat M,\bar M,\bar M_i,L,E,\dots,K,\hat\rho,\hat E$ as in Definition 3.2, and in addition*

(S1) $\hat c\hat E<1$ *(so $\hat E<2^{-13}/n$);*
(S2) $L\ge\max\{\bar M,M^{-1},\hat M^{-1}\}$ *(this is the definition; it is stated to allow L larger, used in the iteration: replace L by any larger number, and then use the same formulas);*
(S3) $\omega_0(D)\subset\mathcal{D}_{\gamma_1,\gamma_2,\tau}$.

Put $\rho_+:=\hat\rho/16$, $s_+:=s/12$, $M_+:=2M$, $\hat M_+:=2\hat M$, $\bar M_+:=2\bar M$, $\bar M_{i+}:=2\bar M_i$, $L_+:=2L$, $E_+:=E^2LM^2/\gamma_1^2$ and let $K_+,\hat\rho_+,\hat E_+$ be given by Definition 3.2 with the $+$ data. Then there are a map Ψ_+, a function h_+ and a set $D_+\subset D_{\hat\rho/16}$, with

(i) $\Psi_+:(D_+)_{\rho_+}\times\mathbb{T}_{\bar s+s_+}\to D_{\hat\rho}\times\mathbb{T}_{\bar s+s}$ *holomorphic, real on reals, $\Psi_+-\mathrm{id}$ periodic, symplectic, and $H_+:=H\circ\Psi_+=h_++f_+$ with $h_+=h+g$, $g=g(I)$ analytic on $D_{\hat\rho/2}$, $\sup_{D_{\hat\rho/2}}|g|\le\frac54E$, and $\|f_+\|_{\rho_+,\bar s+s_+}\le E_+$;*
(ii) $\omega_+=\partial h_+$ *is injective on $D_{\hat\rho/8}$ with $U_+=\partial^2h_+$ nonsingular there; $(D_+)_{\rho_+}\subset D_{\hat\rho/8}$, $\omega_+(D_+)=\omega_0(D)$ (so (S3) holds again);*
(iii) $M_+\ge\sup_{D_{\hat\rho/4}}\|U_+\|$, $\hat M_+\ge\sup\|\hat U_+\|$, $\bar M_+\ge\sup\|U_+^{-1}\|$, $\bar M_{i+}\ge\sup\|T_{i+}\|$ *(sup over $D_{\hat\rho/4}$), and $L_+\ge\max\{\bar M_+,M_+^{-1},\hat M_+^{-1}\}$;*
(iv) the map $\hat\imath_+:=(\omega_+|_{D_{\hat\rho/8}})^{-1}\circ\omega_0$ is defined and holomorphic on $D_{\hat\rho/16}$, $D_+=\hat\imath_+(D)$, and

$$\sup_{D_{\hat\rho/16}}|\hat\imath_{+,i}-\mathrm{id}_i|\le 10\frac{\bar M_i}{\bar M}\hat E\hat\rho\ (\le 10\hat E\hat\rho,\ i=1,2),\qquad L_D(\hat\imath_+-\mathrm{id})\le 640\,n\hat E;$$

(where ω_0 here means the frequency map of the current h, and L_D is the Lipschitz constant on the set D);
(v) the displacement bounds, for $(I',\theta')\in D_{\hat\rho/2}\times\mathbb{T}_{\bar s+s/6}$ and $\Psi_+(I',\theta')=(I,\theta)$:

$$|I_1-I_1'|\le\tfrac34\tfrac{\hat M}{\bar M}\hat E\hat\rho,\quad |I_2-I_2'|\le\tfrac34\hat E\hat\rho,\quad 2\pi|\theta-\theta'|\le\tfrac34\hat Es;$$

(vi) $\hat\rho_+\ge\hat\rho/(2\cdot24^{\tau+1})$, $K_+\le 24K$, $\hat E_+\le\frac89(4s)^{2(\tau+1)}\hat E^2<\hat E^2$, *and* $\hat c\hat E_+<1$.

Proof. **Step S3.14: Non-resonance on $D_{\hat\rho}$.** Let $I\in D_{\hat\rho}$, pick $I_0\in D$ with $|I-I_0|<\hat\rho\le\rho$; the segment $[I_0,I]$ lies in D_ρ, where $\|U\|\le M$, hence $|\omega_i(I)-\omega_i(I_0)|\le M\hat\rho$ for all i, and for $i>n_1$ (rows of $\hat U$) $\le\hat M\hat\rho$ ([K:bp:p1:ext-mvt]). For $0<|k|_1\le K$ and $k_1\neq0$: $|k\cdot\omega(I)|\ge\gamma_1|k|_1^{-\tau}-|k|_1M\hat\rho\ge\gamma_1K^{-\tau}-KM\hat\rho\ge\frac23\gamma_1K^{-\tau}=:\alpha_1$, since $\hat\rho\le\gamma_1/(3MK^{\tau+1})$ and $\omega(I_0)\in\mathcal{D}_{\gamma_1,\gamma_2,\tau}$ by (S3). For $k_1=0$ only ω_2 enters: $|k\cdot\omega(I)|\ge\gamma_2K^{-\tau}-K\hat M\hat\rho\ge\frac23\gamma_2K^{-\tau}=:\alpha_2$. Also $\alpha_2\le\alpha_1$ and (from $\hat\rho\le\gamma_i/(3\cdot K^{\tau+1}\cdot(\mathrm{M}\text{ or }\hat M))$)

$$\alpha_1\ge 2MK\hat\rho,\qquad \alpha_2\ge 2\hat MK\hat\rho. \tag{3.2}$$

Step S3.15: Smallness and $Ks \geq 6$. $Ks = 6\log_+(x^{-1}) \geq 6$. By (3.2) and $L \geq \hat{M}^{-1}$: $64e\,KE/(\alpha_2\hat{\rho}) \leq 64e\,E/(2\hat{M}\hat{\rho}^2) \leq 32e\,\hat{E} < 87\hat{E} < \hat{c}\hat{E} < 1$. So Lemma 3.8 applies with $r = \hat{\rho}$.

Step S3.16: Averaging. Lemma 3.8 gives Ψ, g, f_* on $D_{\hat{\rho}/2}\times\mathbb{T}_{\bar{s}+s/6}$. Since $e^{-Ks/6} = \min\{e^{-1}, x\} \leq x$ ($Ks/6 = \log_+(1/x)$), $\|f_*\| \leq xE = E^2LM^2/\gamma_1^2 = E_+$. Define Ψ_+ as the restriction of Ψ to $(D_+)_{\rho_+}\times\mathbb{T}_{\bar{s}+s_+}$ once D_+ is defined below ($(D_+)_{\rho_+} \subset D_{\hat{\rho}/8} \subset D_{\hat{\rho}/2}$ and $s_+ \leq s/6$; the norm is monotone). Also $\sup_{D_{\hat{\rho}/2}}|g| \leq \sup|g - f_0| + \sup|f_0| \leq E/4 + E$. This proves the first part of (i).

Step S3.17: Displacements. By the last line of Lemma 3.8, (3.2), $Ks \geq 6$ and $M\hat{M}^{-1}$-bounds: $|I_1 - I_1'| \leq \frac{9E}{\alpha_1 s} \leq \frac{9EK}{6\alpha_1} \leq \frac{9E}{12M\hat{\rho}} = \frac{3}{4}\frac{E}{M\hat{\rho}} \leq \frac{3}{4}\frac{\hat{M}}{M}\frac{EL}{\hat{\rho}}$, using $\hat{M}L \geq 1$; and $\frac{EL}{\hat{\rho}} = \hat{E}\hat{\rho}$. Likewise $|I_2 - I_2'| \leq \frac{9E}{\alpha_2 s} \leq \frac{3}{4}\frac{E}{\hat{M}\hat{\rho}} \leq \frac{3}{4}\hat{E}\hat{\rho}$, $2\pi|\theta - \theta'| \leq \frac{9E}{\alpha_2\hat{\rho}} \leq \frac{9E}{2\hat{M}K\hat{\rho}^2} \leq \frac{9}{2}\frac{\hat{E}}{K} \leq \frac{3}{4}\hat{E}s$ (as $K \geq 6/s$). This is (v).

Step S3.18: Derivatives of g. For $I \in D_{\hat{\rho}/4}$ the polydisc of radius $\hat{\rho}/4$ around I lies in $D_{\hat{\rho}/2}$. [K:`bp:p1:cauchy-open`] with $A = \frac{5}{4}E$, $\delta = \hat{\rho}/4$ gives $|\partial_i g(I)| \leq 5E/\hat{\rho}$ and $|\partial_i\partial_j g(I)| \leq 4A/\delta^2 = 80E/\hat{\rho}^2$, so

$$\sup_{D_{\hat{\rho}/4}}\|\partial^2 g\| \leq \frac{80nE}{\hat{\rho}^2} = \frac{80n\hat{E}}{L}, \qquad \sup_{D_{\hat{\rho}/4}}|\partial g| \leq \frac{5E}{\hat{\rho}}. \tag{3.3}$$

Since $L^{-1} \leq M$, $L^{-1} \leq \hat{M}$ and $\bar{M} \leq L$: $\|\partial^2 g\| \leq 80n\hat{E}\min\{M, \hat{M}\} \leq \frac{1}{100}\min\{M, \hat{M}\}$ and $\bar{M}\|\partial^2 g\| \leq 80n\hat{E} \leq \frac{1}{100}$.

Step S3.19: Bounds on U_+. $U_+ = U + \partial^2 g$ on $D_{\hat{\rho}/4}$ and $\hat{U}_+ = \hat{U} + (\partial^2 g)_{\mathrm{rows}>\mathrm{n}_1}$ with $\|\mathrm{rows}\| \leq \|\partial^2 g\|$. Hence $\|U_+\| \leq M + \frac{M}{100} \leq 2M$ and $\|\hat{U}_+\| \leq \hat{M} + \frac{\hat{M}}{100} \leq 2\hat{M}$. Further $U_+ = (1 + \partial^2 g\,U^{-1})U$, so $U_+^{-1} = U^{-1}(1 + \partial^2 g\,U^{-1})^{-1}$ ($\|\partial^2 g\,U^{-1}\| \leq 1/100$, (X5)), hence $\|U_+^{-1}\| \leq 2\bar{M}$ and, the rows of a product T_iB being T_i times B, $\|T_{i+}\| \leq 2\|T_i\| \leq 2\bar{M}_i$ (the left factor of $U^{-1}\cdot(\ldots)$ keeps row blocks). Nonsingular. And $L_+ = 2L \geq \max\{2\bar{M}, (2M)^{-1}, (2\hat{M})^{-1}\}$. This is (iii) on $D_{\hat{\rho}/4}$.

Step S3.20: Injectivity of ω_+ on $D_{\hat{\rho}/8}$. Let $I, I' \in D_{\hat{\rho}/8}$ with $\omega_+(I) = \omega_+(I')$ and put $w = \omega_0(I)$, $w' = \omega_0(I')$. Then $w - w' = \partial g(I') - \partial g(I)$ and $|w - w'| \leq 10E/\hat{\rho}$ by (3.3). The ball $B(I, \hat{\rho}/8) \subset D_{\hat{\rho}/4} \subset D_\rho$, and ω_0 is injective on D_ρ with $\|U^{-1}\| \leq \bar{M}$; Lemma 3.6 gives $\omega_0(B(I, \hat{\rho}/8)) \supset B(w, \hat{\rho}/(8\bar{M}))$. Since $10E/\hat{\rho} < \hat{\rho}/(8\bar{M})$ (equivalent to $80E\bar{M} < \hat{\rho}^2$, implied by $80\hat{E} < 1$ and $\bar{M} \leq L$), $w' = \omega_0(I'')$ with $I'' \in B(I, \hat{\rho}/8)$ and, by injectivity of ω_0 on D_ρ, $I'' = I'$. Thus the segment $[I, I'] \subset D_{\hat{\rho}/4}$ and, by Lemma 3.6 (second part) and the mean value inequality for ∂g along $[I, I']$, $|I - I'| \leq \bar{M}|w - w'| \leq \bar{M}\sup\|\partial^2 g\||I - I'| \leq 80n\hat{E}|I - I'|$, whence $I = I'$. With (iii), ω_+ is a holomorphic injection with nonsingular derivative on $D_{\hat{\rho}/8}$; by (X1) an open map with holomorphic inverse, $\|D(\omega_+^{-1})\| \leq 2\bar{M}$ (and $\|(D\omega_+^{-1})_i\| \leq 2\bar{M}_i$ for the row blocks) on $\omega_+(D_{\hat{\rho}/8})$.

Step S3.21: The map $\hat{\imath}_+$ and D_+. For $I \in D_{\hat{\rho}/16}$, $B(I, \hat{\rho}/16) \subset D_{\hat{\rho}/8}$ and Lemma 3.6 for $F = \omega_+$, $m = 2\bar{M}$ gives $\omega_+(B(I, \hat{\rho}/16)) \supset B(\omega_+(I), \hat{\rho}/(32\bar{M}))$. But $\omega_0(I) = \omega_+(I) - \partial g(I)$ and $|\partial g(I)| \leq 5E/\hat{\rho} < \hat{\rho}/(32\bar{M})$ (as $160E\bar{M} < \hat{\rho}^2$, because $160\hat{E} < 1$). So $\hat{\imath}_+ := \omega_+^{-1}\circ\omega_0$ is defined and holomorphic on $D_{\hat{\rho}/16}$, with values in $B(I, \hat{\rho}/16)$, $\omega_+\circ\hat{\imath}_+ = \omega_0$. By Lemma 3.6 (second part, with $w_1 = \omega_0(I)$, $w_2 = \omega_+(I)$): $|\hat{\imath}_+(I) - I| \leq 2\bar{M}|\partial g(I)| \leq 10\bar{M}E/\hat{\rho}$, and the same estimate holds for the first n_1 and last n_2 components with $\bar{M}_i$ instead of $\bar{M}$ (use the row blocks of $D\omega_+^{-1}$). By $\bar{M} \leq L$: $10\bar{M}_iE/\hat{\rho} \leq 10\frac{\bar{M}_i}{\bar{M}}\frac{EL}{\hat{\rho}} = 10\frac{\bar{M}_i}{\bar{M}}\hat{E}\hat{\rho} \leq 10\hat{E}\hat{\rho}$. This is the sup bound in (iv). Then $\sup|\hat{\imath}_+ - \mathrm{id}| \leq 10\hat{E}\hat{\rho} < \hat{\rho}/16$ since $\hat{E} < 2^{-13}$. Lipschitz bound: Lemma 3.7 with $S = D$, $r = \hat{\rho}/16$, $N = n$, $A = 10\hat{E}\hat{\rho}$ gives $L_D(\hat{\imath}_+ - \mathrm{id}) \leq 4n\cdot 10\hat{E}\hat{\rho}\cdot 16/\hat{\rho} = 640n\hat{E}$. Since h is real analytic, ω_0 and ω_+ are real on $\mathbb{R}^n$; if $\omega_+(I'') = \omega_0(I)$ with $I \in D \subset \mathbb{R}^n$, then $\overline{I''}$ is a second preimage in the same ball, so $I'' = \overline{I''} \in \mathbb{R}^n$ by injectivity. Hence $\hat{\imath}_+(D) \subset \mathbb{R}^n$. Put $D_+ := \hat{\imath}_+(D) \subset D_{\hat{\rho}/16} \cap \mathbb{R}^n$. Since $\omega_+\circ\hat{\imath}_+ = \omega_0$ on D and ω_0 is injective, $\hat{\imath}_+$ is injective on D and

$\omega_+(D_+)=\omega_0(D)$. Then $(D_+)_{\rho_+}\subset D_{\hat\rho/16+\hat\rho/16}=D_{\hat\rho/8}$. This is (ii) and the rest of (iv); in particular $\omega_+(D_+)=\omega_0(D)\subset\mathcal{D}_{\gamma_1,\gamma_2,\tau}$.

Step S3.22: The new constants. Put $x=EM^2L/\gamma_1^2$; then $E_+=Ex$ and $x_+:=E_+L_+M_+^2/\gamma_1^2=Ex\cdot 2L\cdot 4M^2/\gamma_1^2=8x^2$. Hence $\log(1/x_+)=2\log(1/x)-3\log 2\le 2\log_+(1/x)$ and $\log_+(1/x_+)\le 2\log_+(1/x)$ (as $1\le\log_+$), so

$$K_+=\frac{6}{s_+}\log_+\frac{1}{x_+}\le\frac{72}{s}\cdot 2\log_+\frac{1}{x}=24K.$$

Therefore $\gamma_1/(3M_+K_+^{\tau+1})\ge\frac12 24^{-(\tau+1)}\gamma_1/(3MK^{\tau+1})$, likewise for $\gamma_2,\hat M$, and $\rho_+=\hat\rho/16$; as $\hat\rho\le\min\{\gamma_1/(3MK^{\tau+1}),\gamma_2/(3\hat MK^{\tau+1})\}$, $\hat\rho_+\ge\hat\rho\min\{\frac{1}{2\cdot 24^{\tau+1}},\frac{1}{16}\}=\hat\rho/(2\cdot 24^{\tau+1})$.

Step S3.23: The quadratic bound. $E_+L_+=2E^2L^2M^2/\gamma_1^2$ and $\hat\rho_+^{-2}\le 4\cdot 24^{2(\tau+1)}\hat\rho^{-2}$, so $\hat E_+\le 8\cdot 24^{2(\tau+1)}\hat E^2\hat\rho^2M^2/\gamma_1^2$ (using $E^2L^2/\hat\rho^2=\hat E^2\hat\rho^2$). From $\hat\rho\le\gamma_1/(3MK^{\tau+1})$ and $K\ge 6/s$: $\hat\rho^2M^2/\gamma_1^2\le\frac19(s/6)^{2(\tau+1)}$. Hence $\hat E_+\le\frac89(24s/6)^{2(\tau+1)}\hat E^2=\frac89(4s)^{2(\tau+1)}\hat E^2<\hat E^2$ since $4s\le\bar s<1$. Finally $\hat c\hat E_+<\hat c\hat E\cdot\hat E<1$. This is (vi). □

Remark 3.10 (Differences from CP10 in the step). CP10 (74)–(75), (86)–(87) have $5n$, 2^6n and 2^7 from a mixture of 1-, ∞- and "sup" norms; the above constants are those of max norms. CP10 injectivity argument (p. 569–570) uses "$\sup\|\partial(\omega^{-1})\|\sup|\omega(I)-\omega(I')|$", a mean value inequality that silently needs $[\omega(I),\omega(I')]\subset\omega(D_\rho)$; Lemma 3.6 supplies it. The Lipschitz bound of $\hat\imath_+-\mathrm{id}$ on the (non-convex) set D is repaired by Lemma 3.7. CP10 (91) prints $(4s)^{\tau+1}$ where the computation gives $(4s)^{2(\tau+1)}$.

3.7. **The iterative lemma.**

Definition 3.11 (The sequences)**.** Given the data and constants of Proposition 3.3, put $D_0=D^{\mathrm{dio}}=\omega_0^{-1}(\mathcal{D}_{\gamma_1,\gamma_2,\tau})\cap D$, $\rho_0=\rho$, $s_0=s$, $M_0=M$, $\hat M_0=\hat M$, $\bar M_0=\bar M$, $\bar M_{i,0}=\bar M_i$, $L_0=L$, $E_0=E$, $h_0=h$, $f_0=f$ and, recursively for $j\ge1$: $s_j=s_{j-1}/12$, $M_j=2M_{j-1}$, $\hat M_j=2\hat M_{j-1}$, $\bar M_j=2\bar M_{j-1}$, $\bar M_{i,j}=2\bar M_{i,j-1}$, $L_j=2L_{j-1}$, $\rho_j=\hat\rho_{j-1}/16$, $E_j=E_{j-1}^2L_{j-1}M_{j-1}^2/\gamma_1^2$, and $K_j,\hat\rho_j,\hat E_j$ by Definition 3.2 with the data $(s_j,\rho_j,E_j,M_j,\hat M_j,\bar M_j)$ (so L_j there is $\max\{\bar M_j,M_j^{-1},\hat M_j^{-1}\}\le 2^jL$, and in the formulas the larger $L_j=2^jL$ is used, as allowed by (S2) of Lemma 3.9).

Lemma 3.12 (Iterative lemma)**.** *Under the hypotheses of Proposition 3.3 (in particular $\hat c\hat E_0<1$), for every $j\ge0$ there are a set $D_j\subset\mathbb{R}^n$, an analytic symplectic map $\Phi_j=\Psi_1\circ\cdots\circ\Psi_j$ on $W_j:=(D_j)_{\rho_j}\times\mathbb{T}_{\bar s+s_j}$ (with $\Phi_0=\mathrm{id}$, D_0 as above) and $H_j=H\circ\Phi_j=h_j+f_j$, such that*

- *(i) $D_j\subset(D_{j-1})_{\hat\rho_{j-1}/16}$, $\omega_j=\partial h_j$ is a diffeomorphism of $(D_j)_{\rho_j}$ onto an open set, $\omega_j(D_j)=\omega_0(D_0)\subset\mathcal{D}_{\gamma_1,\gamma_2,\tau}$; the map $\hat\imath_j:=\omega_j^{-1}\circ\omega_{j-1}:D_{j-1}\to D_j$ (defined and holomorphic on $(D_{j-1})_{\hat\rho_{j-1}/16}$) satisfies $\sup|\hat\imath_{j,i}-\mathrm{id}_i|\le 10\frac{\bar M_i}{M}\hat E_{j-1}\hat\rho_{j-1}$ and $L_{D_{j-1}}(\hat\imath_j-\mathrm{id})\le 640n\hat E_{j-1}$;*
- *(ii) $\|f_j\|_{\rho_j,\bar s+s_j}\le E_j$;*
- *(iii) $\Psi_j:W_j\to W'_{j-1}$ with $W'_{j-1}:=D_{\hat\rho_{j-1}}\times\mathbb{T}_{\bar s+s_{j-1}}\supset W_{j-1}$ is holomorphic, real on reals, $\Psi_j-\mathrm{id}$ periodic, and on W_j*

$$|\Delta I_1|\le\tfrac34\tfrac{\hat M}{M}\hat E_{j-1}\hat\rho_{j-1},\quad |\Delta I_2|\le\tfrac34\hat E_{j-1}\hat\rho_{j-1},\quad 2\pi|\Delta\theta|\le\tfrac34\hat E_{j-1}s_{j-1};$$

- *(iv) $\hat E_j<\hat E_{j-1}^2$ for $j\ge1$, whence $\hat E_j\le\hat E_0^{2^j}$, and $\hat c\hat E_j<1$;*
- *(v) $\hat\rho_j\le\hat\rho_{j-1}/16$ and $\hat\rho_j\ge\hat\rho_{j-1}/(2\cdot 24^{\tau+1})$, $s_j=s/12^j$, $K_j\le 24^jK$.*

Proof. **Step S3.24: Induction hypothesis.** We prove by induction on $j\ge0$ the statement $\mathcal{P}_j$: the data $(D_j,h_j,f_j,\rho_j,s_j,M_j,\hat M_j,\bar M_j,\bar M_{i,j},L_j,E_j)$ satisfy the hypotheses of Lemma 3.9 (on the domain $(D_j)_{\rho_j}\times\mathbb{T}_{\bar s+s_j}$): $\hat c\hat E_j<1$, $L_j\ge\max\{\bar M_j,M_j^{-1},\hat M_j^{-1}\}$, $\omega_j(D_j)\subset\mathcal{D}_{\gamma_1,\gamma_2,\tau}$, bounds $M_j,\hat M_j,\bar M_j,\bar M_{i,j}$

on $(D_j)_{\rho_j}$, ω_j injective on $(D_j)_{\rho_j}$ with nonsingular U_j, $\|f_j\| \le E_j$, $\hat{M}_j \le M_j$, $\bar{M}_{i,j} \le \bar{M}_j$, and $4s_j \le \bar{s} < 1$.

Step S3.25: Base. $\mathcal{P}_0$ is the hypothesis of Proposition 3.3 and $\omega_0(D_0) \subset \mathcal{D}_{\gamma_1,\gamma_2,\tau}$ by definition of D_0 (note $D_0 \subset D$, so the sup bounds on D_ρ give those on $(D_0)_\rho$; injectivity is inherited).

Step S3.26: Induction. Apply Lemma 3.9 to $\mathcal{P}_{j-1}$: it produces $D_j := D_+$, $\Psi_j := \Psi_+$, $h_j := h_+$, $f_j := f_+$, and (i)–(vi) of that lemma are (i)–(iii) here and the recursion of Definition 3.11 (the quantities named $\hat{\rho}_+, K_+, \hat{E}_+$ there are $\hat{\rho}_j, K_j, \hat{E}_j$). Items (iii) and (v) of the step give $\mathcal{P}_j$ ($\hat{M}_j \le M_j$, $\bar{M}_{i,j} \le \bar{M}_j$ propagate by doubling).

Step S3.27: Composition. $\Phi_j = \Phi_{j-1} \circ \Psi_j$ is defined on W_j because $\Psi_j(W_j) \subset W'_{j-1}$ and in fact $\subset W_{j-1}$: the displacement is $\le \frac{3}{4}\hat{E}_{j-1}\hat{\rho}_{j-1} < \hat{\rho}_{j-1}/16$ and $W_j \subset D_{\hat{\rho}_{j-1}/8} \times \dots$, so $\Psi_j(W_j) \subset D_{\hat{\rho}_{j-1}/8+\hat{\rho}_{j-1}/16} \times \mathbb{T}_{\bar{s}+s_{j-1}}$; this lies in $W_{j-1} = (D_{j-1})_{\rho_{j-1}} \times \mathbb{T}$ because $\hat{\rho}_{j-1} \le \rho_{j-1}$ (definition of $\hat{\rho}$). The identity $H_j = H_{j-1} \circ \Psi_j = H \circ \Phi_j$ is then immediate, and Φ_j is symplectic, real, with $\Phi_j - \mathrm{id}$ periodic.

Step S3.28: (iv)–(v) of the Lemma. By Step (vi) of Lemma 3.9: $\hat{E}_j < \hat{E}_{j-1}^2$, $\hat{\rho}_j \ge \hat{\rho}_{j-1}/(2 \cdot 24^{\tau+1})$, $\hat{\rho}_j \le \rho_j = \hat{\rho}_{j-1}/16$, $K_j \le 24K_{j-1}$. □

4. Two-scale KAM theorem II: the limit, the tori and the measure

We continue with the notation of Definition 3.11 and Lemma 3.12; $c := 24^{\tau+1}/6$ and $q := 2 \cdot 24^{\tau+1} = 12c$. Since $\hat{c} \ge 2c$, $\hat{c} \ge 2^{13}n$, the KAM condition gives

$$\hat{E}_0 < \frac{1}{8192\, n}, \qquad c\hat{E}_0 < \frac{1}{2}. \tag{3.4}$$

Lemma 4.1 (Sums). *For $m \ge 0$: $\hat{E}_k \le \hat{E}_m^{2^{k-m}}$ for $k \ge m$, $\sum_{k\ge m} \hat{E}_k \le 2\hat{E}_m$, and $\sum_{k\ge 0} c^k \hat{E}_k \le 2\hat{E}_0$. Also $\hat{\rho}_k \le \hat{\rho}_0\, 16^{-k}$, $s_k = s\, 12^{-k}$.*

Proof. **Step S4.1: Doubling.** By Lemma 3.12(iv), $\hat{E}_k < \hat{E}_{k-1}^2$, so $\hat{E}_k \le \hat{E}_m^{2^{k-m}} \le \hat{E}_m^{k-m+1}$ (as $2^i \ge i+1$ and $\hat{E}_m < 1$). Hence $\sum_{k\ge m} \hat{E}_k \le \hat{E}_m/(1-\hat{E}_m) \le 2\hat{E}_m$.

Step S4.2: Weighted sum. $c^k\hat{E}_k \le c^k\hat{E}_0^{k+1} = \hat{E}_0(c\hat{E}_0)^k \le \hat{E}_0 2^{-k}$ by (3.4); sum $\le 2\hat{E}_0$.

Step S4.3: Radii. $\hat{\rho}_k \le \rho_k = \hat{\rho}_{k-1}/16$ (Definition of $\hat{\rho}_k$), iterate. □

4.1. The limit actions (CP10 Step 1).

Lemma 4.2 (Limit actions). *Put $\check{\imath}_0 = \mathrm{id}$, $\check{\imath}_j := \hat{\imath}_j \circ \dots \circ \hat{\imath}_1 : D_0 \to D_j$. Then $\check{\imath}_j$ converges uniformly on D_0 to $\check{\imath} = (\check{\imath}_1, \check{\imath}_2)$ and*

- *(i) $\sup_{D_0}|\check{\imath}_i - \mathrm{id}_i| \le 20\frac{\bar{M}_i}{\bar{M}}\hat{E}_0\hat{\rho}_0 \le 20\hat{E}_0\hat{\rho}_0$; $\sup_{D_0}|\check{\imath} - \check{\imath}_j| \le 20\hat{\rho}_j\hat{E}_j$;*
- *(ii) $D_* := \check{\imath}(D_0) \subset (D_j)_{\rho_j/2}$ for all j, in particular $D_* \subset \mathrm{Re}(D_0)_{20\hat{E}_0\hat{\rho}_0}$;*
- *(iii) $\lambda := L_{D_0}(\check{\imath} - \mathrm{id}) \le 2560\, n\hat{E}_0 < \frac{1}{3}$; so $\check{\imath}$ is injective, $(1-\lambda)|x-y| \le |\check{\imath}x - \check{\imath}y| \le (1+\lambda)|x-y|$;*
- *(iv) $\omega_j(\check{\imath}_j(I_0)) = \omega_0(I_0)$ for all $I_0 \in D_0$, $j \ge 0$.*

Source: CP10 Step 1, pp. 570–571, (97)–(104); constants recomputed (CP10: $10n$, $2^8 n\hat{E}$).

Proof. **Step S4.4: Well-posedness.** $\hat{\imath}_j : D_{j-1} \to D_j$ (Lemma 3.12(i)); so $\check{\imath}_j(D_0) \subset D_j$ and the composition is defined. (iv) follows from $\omega_j \circ \hat{\imath}_j = \omega_{j-1}$ by induction.

Step S4.5: Telescoping. $|\check{\imath}_{j,i}(x) - \check{\imath}_{j-1,i}(x)| = |\hat{\imath}_{j,i}(y) - y_i|$ with $y = \check{\imath}_{j-1}(x) \in D_{j-1}$, so $\le 10\frac{\bar{M}_i}{\bar{M}}\hat{E}_{j-1}\hat{\rho}_{j-1}$ (the ratio $\bar{M}_{i,j}/\bar{M}_j = \bar{M}_i/\bar{M}$ is constant). Since $\hat{\rho}_{j-1} \le \hat{\rho}_0$, Lemma 4.1 gives the Cauchy property and (i): $\sum_{j\ge1} 10\hat{E}_{j-1}\hat{\rho}_{j-1} \le 20\hat{E}_0\hat{\rho}_0$; the tail is $\le 10\hat{\rho}_j \sum_{m\ge j}\hat{E}_m \le 20\hat{\rho}_j\hat{E}_j$.

Step S4.6: Location. $\check\imath_j(D_0) \subset D_j \subset \mathbb{R}^n$ and $20\hat\rho_j\hat E_j \le \rho_j/2$ ($\hat\rho_j \le \rho_j$, $\hat E_j < 1/40$), so $D_* \subset \mathrm{Re}(D_j)_{\rho_j/2}$; for $j=0$ and $D_0 \subset D$ this gives the last claim of (ii).

Step S4.7: Lipschitz constant. Put $i_j = \check\imath_j - \mathrm{id}$. From $\check\imath_{j+1} = \hat\imath_{j+1}\circ\check\imath_j$: $i_{j+1} = (\hat\imath_{j+1}-\mathrm{id})\circ\check\imath_j + i_j$, so $1+L(i_{j+1}) \le (1+L(\hat\imath_{j+1}-\mathrm{id}))(1+L(i_j))$ and, by Lemma 3.12(i), $1+L(i_j) \le \prod_{k\le j}(1+640n\hat E_{k-1}) \le \exp(640n\sum_k \hat E_{k-1}) \le \exp(1280n\hat E_0)$. As $x := 1280n\hat E_0 < \frac16$, $e^x \le 1+2x$: $L(i_j) \le 2560n\hat E_0$ uniformly in j; the pointwise limit of Lipschitz maps keeps the constant. Injectivity and the two-sided bound follow from $|\check\imath x - \check\imath y| \ge |x-y| - \lambda|x-y|$. □

4.2. **The limit symplectic map (CP10 Step 2).** *Rescaling.* Let $S(I,\theta) := (I/\hat\rho_0,\ 2\pi\theta/s_0)$ and $|(I,\theta)|_* := |S(I,\theta)|$ (max norm); for a map G put $\check G = SGS^{-1}$ and L_* for Lipschitz constants in $|\cdot|_*$. Put $\mathcal{D}_j := \mathrm{Re}(D_j)_{\rho_j/2}$. All maps below are real on real points.

Lemma 4.3 (Domains and rescaled Lipschitz bounds for Ψ_k)**.** *(i) $\Psi_k(\mathcal{D}_k\times\mathbb{R}^n) \subset \mathcal{D}_{k-1}\times\mathbb{R}^n$ ($k\ge1$); hence $\Phi_j = \Psi_1\circ\cdots\circ\Psi_j$ is defined on $\mathcal{D}_j\times\mathbb{R}^n$ with values in $\mathcal{D}_0\times\mathbb{R}^n$.*
(ii) $L_*(\check\Psi_k - \mathrm{id}; \mathcal{D}_k\times\mathbb{R}^n) \le 192\,n\,c^{k-1}\hat E_{k-1} =: \varsigma_k$, and $\sum_{k\ge1}\varsigma_k \le 384n\hat E_0 < \frac{1}{20}$.
(iii) $|\Psi_k(y) - y|_* \le \frac34\hat E_{k-1}12^{-(k-1)} \le \frac34\hat E_{k-1}$ for $y\in\mathcal{D}_k\times\mathbb{R}^n$.

Proof. **Step S4.8: Domains.** Let $y=(I,\theta)$, $I\in\mathcal{D}_k$: $\mathrm{dist}(I,D_k) < \rho_k/2 = \hat\rho_{k-1}/32$ and $D_k \subset \mathrm{Re}(D_{k-1})_{\hat\rho_{k-1}/16}$. By Lemma 3.12(iii) the real displacement is $\le \frac34\hat E_{k-1}\hat\rho_{k-1}$. Hence $\Psi_k(y)_I$ is within $\hat\rho_{k-1}(\frac{1}{32}+\frac{1}{16}+\frac34\hat E_{k-1}) < \hat\rho_{k-1}/8$ of D_{k-1}; and $\hat\rho_{k-1}/8 < \rho_{k-1}/2$ (for $k=1$ because $\hat\rho_0\le\rho$; for $k\ge2$ because $\hat\rho_{k-1}\le\hat\rho_{k-2}/16$).

Step S4.9: Bound on displacement. Scaled: $|\Delta I|/\hat\rho_0 \le \frac34\hat E_{k-1}\hat\rho_{k-1}/\hat\rho_0 \le \frac34\hat E_{k-1}16^{-(k-1)}$ (Lemma 4.1; $\hat M \le M$) and $2\pi|\Delta\theta|/s_0 \le \frac34\hat E_{k-1}s_{k-1}/s_0 = \frac34\hat E_{k-1}12^{-(k-1)}$. Thus $A_k := \sup_{W_k}|\check\Psi_k - \mathrm{id}| \le \frac34\hat E_{k-1}12^{-(k-1)}$; this is (iii).

Step S4.10: Lipschitz. Apply Lemma 3.7 in the scaled coordinates, $N=2n$, $S = S(\mathcal{D}_k\times\mathbb{R}^n)$, radius $\varepsilon_k = \min\{\rho_k/(2\hat\rho_0),\ s_k/s_0\}$: the scaled complex ε_k-neighbourhood of a real point of S lies in $(D_k)_{\rho_k}\times\mathbb{T}_{s_k} \subset W_k$ ($\varepsilon_k\hat\rho_0 \le \rho_k/2$ and $\varepsilon_k s_0 \le s_k \le \bar s + s_k$), where the bound A_k of (iii) holds (Lemma 3.12(iii) holds on the complex W_k). Now $\rho_k/(2\hat\rho_0) = \hat\rho_{k-1}/(32\hat\rho_0) \ge q^{-(k-1)}/32$ (Lemma 3.12(v)) and $s_k/s_0 = 12^{-k} \ge q^{-(k-1)}/32$, so $\varepsilon_k \ge q^{-(k-1)}/32$ and

$$L_* \le \frac{4\cdot 2n\,A_k}{\varepsilon_k} \le 8n\cdot\tfrac34\hat E_{k-1}12^{-(k-1)}\cdot 32q^{k-1} = 192n\,(q/12)^{k-1}\hat E_{k-1} = 192n\,c^{k-1}\hat E_{k-1}.$$

Step S4.11: Sum. Lemma 4.1: $\sum_k\varsigma_k = 192n\sum_{m\ge0}c^m\hat E_m \le 384n\hat E_0 < 384/8192 < \frac{1}{20}$. □

Lemma 4.4 (The limit map Φ)**.** *On $W_* := D_*\times\mathbb{R}^n$ the maps Φ_j converge uniformly to a map Φ with $\Phi - \mathrm{id}$ $\mathbb{Z}^n$-periodic in θ, real analytic in θ, and*

(i) $\sup_{W_*}|\Pi_{I_1}\Phi - \mathrm{id}| \le \frac32\frac{\hat M}{M}\hat E_0\hat\rho_0$, $\sup|\Pi_{I_2}\Phi - \mathrm{id}| \le \frac32\hat E_0\hat\rho_0$, $\sup 2\pi|\Pi_\theta\Phi - \mathrm{id}| \le \frac32\hat E_0 s_0$, and the same bounds on complex θ with $|\mathrm{Im}\,\theta| < \bar s/2\pi$;
(ii) $L_*(\check\Phi - \mathrm{id}; W_*) \le 768\,n\hat E_0 < \frac{1}{10}$; hence Φ is injective on W_* and $|x-y|_*\cdot\frac{9}{10} \le |\Phi x - \Phi y|_* \le \frac{11}{10}|x-y|_*$;
(iii) $\Phi(I,\theta+k) = \Phi(I,\theta) + (0,k)$ for $k\in\mathbb{Z}^n$.

Source: CP10 Step 2, pp. 571–572, (105)–(108).

Proof. **Step S4.12: Telescoping of the displacements.** Write $(I^{(k)},\theta^{(k)}) = \Psi_{k+1}\circ\cdots\circ\Psi_j(x)$ for $x\in W_* \subset W_j$ (Lemma 4.2(ii)); then $\Pi_I\Phi_j(x) - \Pi_I x = \sum_{k=1}^j(I^{(k-1)} - I^{(k)})$ and every summand is bounded by Lemma 3.12(iii). Summing with Lemma 4.1 ($\hat\rho_{k-1}\le\hat\rho_0$, $s_{k-1}\le s_0$, $\sum\hat E_{k-1} \le 2\hat E_0$) gives (i), uniformly in j and also for complex θ (Ψ_k is defined on the complex strip).

Step S4.13: Cauchy sequence. For $x \in W_*$: $\Phi_j(x) - \Phi_{j-1}(x) = \Phi_{j-1}(\Psi_j x) - \Phi_{j-1}(x)$, with $x \in \mathcal{D}_j \times \mathbb{R}^n$ and $\Psi_j x \in \mathcal{D}_{j-1} \times \mathbb{R}^n$ (Lemma 4.3(i)). By the composition rule $L_*(\check{\Phi}_{j-1} - \mathrm{id}) \le \prod_{k<j}(1+\varsigma_k) - 1 \le e^{\sum \varsigma_k} - 1 \le 2\sum \varsigma_k \le \frac{1}{10}$ (using $e^x \le 1 + 2x$, $x \le 1$; the composition formula is that of Lemma 4.2, proof step S4.7, valid because each Ψ_k maps its domain into the next one). Therefore $|\Phi_j(x) - \Phi_{j-1}(x)|_* \le \frac{11}{10}|\Psi_j x - x|_* \le \frac{11}{10} \cdot \frac{3}{4}\hat{E}_{j-1}$, which is summable: uniform convergence on W_*. The limit satisfies (ii) with $L_* \le 2\sum \varsigma_k \le 768 n\hat{E}_0$ (the constant passes to the pointwise limit).

Step S4.14: Periodicity and injectivity. Each $\Psi_k - \mathrm{id}$ is $\mathbb{Z}^n$-periodic in θ (Lemma 3.12), so (iii) holds for Φ_j and in the limit. From (ii), $|\Phi x - \Phi y|_* \ge \frac{9}{10}|x - y|_*$.

Step S4.15: Analyticity in θ. Fix $I_* \in D_*$; $\vartheta \mapsto \Phi_j(I_*, \vartheta) - (0, \vartheta)$ is holomorphic on $\{|\mathrm{Im}\,\vartheta| < \bar{s}/2\pi\} \subset \mathbb{C}^n$ (as $I_* \in (D_j)_{\rho_j}$), bounded uniformly in j by (i), and converges at every real ϑ. By (X2) it converges locally uniformly together with all derivatives to a holomorphic function; its real restriction is $\Phi(I_*, \cdot) - (0, \mathrm{id})$ and the bounds (i) pass to the limit. □

4.3. **The tori (CP10 Steps 2–3 and the estimates (35)).** Define, for $\omega \in \Omega_* = \omega_0(D) \cap \mathcal{D}_{\gamma_1,\gamma_2,\tau}$, $I_0(\omega) := \omega_0^{-1}(\omega) \in D_0$ (injectivity of ω_0; $\omega \in \omega_0(D) \cap \mathcal{D}$ means $I_0 \in D^{\mathrm{dio}} = D_0$), $I_*(\omega) := \check{\imath}(I_0(\omega)) \in D_*$ and

$$\varphi_\omega(\vartheta) := \Phi(I_*(\omega), \vartheta) = (v(\vartheta;\omega),\ \vartheta + u(\vartheta;\omega)).$$

Lemma 4.5 (Estimates and location). *For $\omega \in \Omega_*$: $|u| \le \frac{3}{4\pi}\hat{E}_0 s$ on $\{|\mathrm{Im}\,\vartheta| < \bar{s}/2\pi\}$; $|v_1 - I_{0,1}| \le 20(\frac{\bar{M}_1}{\bar{M}} + \frac{\hat{M}}{M})\hat{E}_0\hat{\rho}_0$, $|v_2 - I_{0,2}| \le 20(\frac{\bar{M}_2}{\bar{M}} + 1)\hat{E}_0\hat{\rho}_0$ on $\mathbb{R}^n$, and $v \in \mathrm{Re}(D_r)$ with $r = 32\hat{E}_0\hat{\rho}_0$, so $\varphi_\omega(\mathbb{R}^n) \subset \mathrm{Re}(D_r) \times \mathbb{R}^n$. This is Proposition 3.3(a), (d), and the location claim.*
Source: CP10 (35), (105)–(107), Step 2 end, p. 572. Misprint: *CP10 (35) bounds $|v_2 - I_2^0|$ by $10n(\bar{M}_2/\bar{M} + \hat{M}/M)\hat{E}\hat{\rho}$ although (107) only gives $2\hat{E}\hat{\rho}$ for I_2 without the factor $\hat{M}/M$.*

Proof. **Step S4.16: Sup bounds.** $|v_i - I_{0,i}| \le |\Pi_{I_i}\Phi(I_*, \vartheta) - I_{*,i}| + |I_{*,i} - I_{0,i}|$, the two terms bounded by Lemma 4.4(i) and Lemma 4.2(i): $\frac{3}{2}\frac{\hat{M}}{M}\hat{E}\hat{\rho} + 20\frac{\bar{M}_1}{\bar{M}}\hat{E}\hat{\rho}$ for $i = 1$ (and $\frac{3}{2} + 20\frac{\bar{M}_2}{\bar{M}}$ for $i = 2$), both $\le$ the stated constants. $|u| \le \frac{3}{2}\hat{E}s/2\pi$ is the angle bound.

Step S4.17: Location. $I_* \in \mathrm{Re}(D_0)_{20\hat{E}\hat{\rho}}$ and $|v - I_*| \le \frac{3}{2}\hat{E}\hat{\rho}$, so $\mathrm{dist}(v, D) < 22\hat{E}\hat{\rho} < r$. □

Lemma 4.6 (Immersion and Lagrangian property). *For $\omega \in \Omega_*$, $\|D_\vartheta u\| \le \frac{3}{4}n\hat{E}_0$ (so $D_\vartheta\varphi_\omega = \binom{D_\vartheta v}{1 + D_\vartheta u}$ has injective lower block $1 + D_\vartheta u$, since $\frac{3}{4}n\hat{E}_0 < \frac{1}{2}$), and $\varphi_\omega^*(\sum_i dJ_i \wedge d\theta_i) = 0$.*

Proof. **Step S4.18: Cauchy in the strip.** u is holomorphic on $|\mathrm{Im}\,\vartheta| < \bar{s}/2\pi$ with $|u| \le 3\hat{E}s/4\pi$. At real ϑ the polydisc of radius $\bar{s}/4\pi$ lies in the strip, so $|\partial_a u_i| \le \frac{3\hat{E}s}{4\pi} \cdot \frac{4\pi}{\bar{s}} = 3\hat{E}\frac{s}{\bar{s}} \le \frac{3}{4}\hat{E}$ ($4s \le \bar{s}$), and the row sum is $\le \frac{3}{4}n\hat{E}$.

Step S4.19: Each Φ_j-torus is Lagrangian. Φ_j is symplectic and $\vartheta \mapsto (I_*, \vartheta)$ has pullback $\sum dI_i \wedge d\theta_i = 0$ (the I-coordinates are constant), so $\varphi^{(j)} := \Phi_j(I_*, \cdot)$ has $\Omega_j(a,b) := \sum_i(\partial_a v_i^{(j)}\partial_b\Theta_i^{(j)} - \partial_b v_i^{(j)}\partial_a\Theta_i^{(j)}) = 0$.

Step S4.20: Limit. By Lemma 4.4 (Step S4.15) the first derivatives of $\varphi^{(j)}$ converge to those of φ_ω (locally uniformly, (X3)); Ω_j is bilinear in them, so $\Omega = \lim \Omega_j = 0$. □

Lemma 4.7 (Invariance). *For $\omega \in \Omega_*$, $\vartheta \in \mathbb{R}^n$, $t \in \mathbb{R}$: $\frac{d}{dt}\varphi_\omega(\vartheta + t\omega) = X_H(\varphi_\omega(\vartheta + t\omega))$, $X_H = (-\partial_\theta H, \partial_J H)$.*
Source: CP10 Step 3, pp. 572–573.

Proof. Let $I_0 = I_0(\omega)$, $I_* = \check{\imath}(I_0)$, $\theta_t = \vartheta + t\omega$, $\gamma(t) = \Phi(I_*, \theta_t)$, $\gamma_j(t) = \Phi_j(I_*, \theta_t)$.

Step S4.21: Symplectic conjugation. Since Φ_j is a symplectic diffeomorphism onto its image and $H_j = H \circ \Phi_j$: $X_H \circ \Phi_j = D\Phi_j \, X_{H_j}$ ([N:bpN:p5:symplectic]). With $X_{H_j}(I_*, \theta) = (-\partial_\theta f_j, \ \omega_j(I_*) + \partial_I f_j)$ we get $\gamma_j'(t) - X_H(\gamma_j(t)) = D\Phi_j(I_*, \theta_t)\big(\partial_\theta f_j, \ \omega - \omega_j(I_*) - \partial_I f_j\big)$ (with $\gamma_j' = D\Phi_j(0, \omega)$).

Step S4.22: $D\Phi_j$ bounded. $\mathcal{D}_j \times \mathbb{R}^n$ is open and $L_*(\check{\Phi}_j) \le \frac{11}{10}$ (Lemma 4.4(ii) argument), so $|S \, D\Phi_j S^{-1}| \le \frac{11}{10}$: $\|D\Phi_j\| \le \kappa := \frac{11}{10} \max\{1, \frac{2\pi\hat{\rho}_0}{s_0}, \frac{s_0}{2\pi\hat{\rho}_0}\}$, independent of j.

Step S4.23: The three error terms. (1) $\omega = \omega_j(\check{\imath}_j(I_0))$ (Lemma 4.2(iv)); $|I_* - \check{\imath}_j(I_0)| \le 20\hat{\rho}_j\hat{E}_j$ and the segment lies in $(D_j)_{\rho_j}$ where $\|U_j\| \le M_j$, so $|\omega - \omega_j(I_*)| \le 20 M_j\hat{\rho}_j\hat{E}_j \le 7\gamma_1\hat{E}_j$, using $M_j\hat{\rho}_j \le \gamma_1/(3K_j^{\tau+1}) \le \gamma_1/3$. (2) $I_* \in \mathrm{Re}(D_j)_{\rho_j/2}$, so the polydisc of radius $\rho_j/2$ about I_* is in $(D_j)_{\rho_j}$ and [K:bp:p1:cauchy-open] gives $|\partial_I f_j| \le 2E_j/\rho_j \le 2E_j/\hat{\rho}_j = 2\hat{E}_j\hat{\rho}_j/L_j \le 2\hat{E}_j M_j\hat{\rho}_j \le \gamma_1\hat{E}_j$ ($L_j \ge M_j^{-1}$). (3) For real θ the polydisc of radius $\bar{s}/4\pi$ in θ lies in $\mathbb{T}_{\bar{s}/2}$, where $|f_j| \le E_j$ (as $\bar{s}/2 \le \bar{s} + s_j$), so $|\partial_\theta f_j| \le 4\pi E_j/\bar{s}$ ([K:bp:p1:cauchy-open]); and $E_j = \hat{E}_j\hat{\rho}_j^2/L_j \le \hat{E}_j\hat{\rho}_j M_j\hat{\rho}_j \le \hat{E}_j\gamma_1\hat{\rho}_0$. All three tend to 0, uniformly in (ϑ, t), as $\hat{E}_j \le \hat{E}_0^{2^j}$. Hence $\sup_t|\gamma_j' - X_H(\gamma_j)| \to 0$.

Step S4.24: Passing to the limit. $\gamma_j \to \gamma$ uniformly in t (Lemma 4.4); X_H is Lipschitz on $\{I \in D_{\rho/2}\} \times \mathbb{T}_{\bar{s}/2}$ ($D\,X_H$ consists of $U + \partial_J^2 f$, $\partial_\theta\partial_J f$, $\partial_\theta^2 f$ and is bounded by $M + C(E, \rho, \bar{s})$ by [K:bp:p1:cauchy-open]) and $\gamma_j(t), \gamma(t)$ lie in its real part ($\gamma(t)$ has action in $\mathrm{Re}(D)_r$, $r \le \rho/2$), so $X_H(\gamma_j) \to X_H(\gamma)$ uniformly. A uniform limit of C^1 functions $\gamma_j \to \gamma$ with $\gamma_j' \to X_H(\gamma)$ uniformly is C^1 with $\gamma' = X_H(\gamma)$. Since $\gamma(t) = \varphi_\omega(\vartheta + t\omega)$, this is the claim. □

4.4. **Measure estimate (CP10 Step 4).** Let $K_\Omega = \bigcup_{\omega\in\Omega_*} \varphi_\omega(\mathbb{R}^n)$; by definition $K_\Omega = \Phi(D_* \times \mathbb{R}^n)$ ($\check{\imath}$ maps $D_0 = D^{\mathrm{dio}}$ onto D_* and $\omega \mapsto I_0(\omega)$ is a bijection $\Omega_* \to D_0$). Put $a := \mathrm{meas}(D \setminus D_0)$, $b := \mathrm{meas}(\mathrm{Re}(D_r) \setminus D)$, $\lambda \le \frac{1}{3}$ as in Lemma 4.2, $\eta := \sup|\check{\imath} - \mathrm{id}| \le 20\hat{E}_0\hat{\rho}_0$.

Lemma 4.8 (Bi-Lipschitz extension and covering)**.** *Let $X \subset \mathbb{R}^N$ be arbitrary, $g : X \to \mathbb{R}^N$ with $L(g) \le \lambda < 1$ in the max norm and $\sup|g| \le \eta$. Then there is a bijection $F = \mathrm{id} + g_e : \mathbb{R}^N \to \mathbb{R}^N$, $F|_X = \mathrm{id} + g$, with $(1-\lambda)|x-y| \le |F(x) - F(y)| \le (1+\lambda)|x-y|$, and for $y \in X$: $F^{-1}(y) \in N_{\eta/(1-\lambda)}(X)$ (real max-norm neighbourhood).*

Proof. **Step S4.25: Extension.** g_e: McShane [K:bp:p1:mcshane], componentwise, with the same λ for the max norm.

Step S4.26: Bijectivity. $|F(x) - F(y)| \ge |x-y| - \lambda|x-y|$ gives injectivity and the lower bound; for $y \in \mathbb{R}^N$ the map $x \mapsto y - g_e(x)$ is a λ-contraction ((X4)), with fixed point $x = F^{-1}(y)$.

Step S4.27: Location of preimages. For $y \in X$ and $x = F^{-1}(y)$: $|x-y| = |g_e(x)| \le |g_e(y)| + \lambda|x-y| \le \eta + \lambda|x-y|$ so $|x-y| \le \eta/(1-\lambda)$. □

Lemma 4.9 (Measure of the complement of D_*)**.** $\mathrm{meas}(D_0 \setminus D_*) \le (4/3)^n(a+b)$ *and hence* $\mathrm{meas}(D \setminus D_*) \le a + (4/3)^n(a+b)$.

Proof. **Step S4.28: Extension and preimage.** Apply Lemma 4.8 with $X = D_0$, $g = \check{\imath} - \mathrm{id}$, $N = n$: $F = \check{\imath}_e$. Let $\rho_1 := \eta/(1-\lambda) \le 30\hat{E}_0\hat{\rho}_0 < r$. If $y \in D_0 \setminus D_*$, then $x = F^{-1}(y) \in N_{\rho_1}(D_0)$ and $x \notin D_0$ (else $y = F(x) = \check{\imath}(x) \in D_*$). So $D_0 \setminus D_* \subset F\big(N_{\rho_1}(D_0) \setminus D_0\big)$.

Step S4.29: Measure. By [K:bp:p1:lip-image] (F is $(1+\lambda)$-Lipschitz), $\mathrm{meas}(D_0 \setminus D_*) \le (1+\lambda)^n \, \mathrm{meas}(N_{\rho_1}(D_0) \setminus D_0)$, $(1+\lambda) \le \frac{4}{3}$. As $D_0 \subset D$: $N_{\rho_1}(D_0) \setminus D_0 \subset (N_{\rho_1}(D) \setminus D) \cup (D \setminus D_0)$, of measure $\le b + a$ ($\rho_1 < r$). The second claim: $D \setminus D_* \subset (D \setminus D_0) \cup (D_0 \setminus D_*)$. □

Lemma 4.10 (Measure of the complement of K_Ω over D)**.** $\mathrm{meas}\big((D \times \mathbb{T}^n) \setminus K_\Omega\big) \le \big[1 + (4/3)^n + 2^n\big(1 + (4/3)^n\big)\big](a+b)$.

Proof. Identify $\mathbb{T}^n$-measure with the unit cell; K_Ω is $\mathbb{Z}^n$-invariant in the angle.

Step S4.30: Splitting. $(D\times\mathbb{T}^n)\setminus K_\Omega \subset \big((D\setminus D_*)\times\mathbb{T}^n\big)\cup\big(((D_*\cap D)\times\mathbb{T}^n)\setminus K_\Omega\big)$. The first set has measure $\le a+(4/3)^n(a+b)$ by Lemma 4.9.

Step S4.31: Rescaled extension. Work in scaled coordinates $x\mapsto Sx$; S is linear, multiplies the Lebesgue measure of every subset of $\mathbb{R}^n_I\times\mathbb{R}^n_\theta$ by the constant $\hat\rho_0^{-n}(2\pi/s_0)^n$ and maps products to products, so all comparisons below between measures of subsets of $\mathbb{R}^n\times\mathbb{R}^n$ are unaffected. By Lemma 4.4, $g=\check\Phi-\mathrm{id}$ on $X=S(D_*\times\mathbb{R}^n)\subset\mathbb{R}^{2n}$ has $L_*(g)\le\lambda':=768n\hat E_0<\frac{1}{10}$ and $\sup|g|\le\eta':=\frac{3}{2}\hat E_0$ (Lemma 4.4(i), scaled). Lemma 4.8 ($N=2n$) gives the bijection $\check\Phi_e=\mathrm{id}+g_e$ of $\mathbb{R}^{2n}$ extending $\check\Phi$ on X, $(1+\lambda')$-Lipschitz, with $|\check\Phi_e^{-1}(y)-y|_*\le\eta'/(1-\lambda')\le 2\hat E_0$ for $y\in X$.

Step S4.32: Preimage structure. Let $y=(I_y,\vartheta_y)$ with $I_y\in D_*\cap D$ and $\vartheta_y\in[0,1)^n$ be a point of the second set, i.e. $y\notin K_\Omega$ (mod $\mathbb{Z}^n$ in the angle), and let $x=(I_x,\vartheta_x):=\Phi_e^{-1}(y)$ (unscaled notation; $y\in D_*\times\mathbb{R}^n=S^{-1}X$). Then $|I_x-I_y|\le\rho_2:=2\hat E_0\hat\rho_0$ and $|\vartheta_x-\vartheta_y|\le\hat E_0 s_0/\pi\le\hat E_0/(4\pi)=:\epsilon$ (as $s_0\le\frac14$). If $I_x\in D_*$, then $x\in D_*\times\mathbb{R}^n$, $\Phi_e(x)=\Phi(x)=y$ and $y\in K_\Omega$, a contradiction. Hence $I_x\in N_{\rho_2}(D_*)\setminus D_*$ and $\vartheta_x\in[-\epsilon,1+\epsilon)^n$.

Step S4.33: Measure. Thus the second set is contained in $\Phi_e\big((N_{\rho_2}(D_*)\setminus D_*)\times[-\epsilon,1+\epsilon)^n\big)$. Φ_e is $(1+\lambda')$-Lipschitz for $|\cdot|_*$ on $\mathbb{R}^{2n}$, so by [K:bp:p1:lip-image] in scaled coordinates (the scaling constants cancel on both sides) its measure is $\le(1+\lambda')^{2n}(1+2\epsilon)^n\,\mathrm{meas}(N_{\rho_2}(D_*)\setminus D_*)\le 2^n\,\mathrm{meas}(N_{\rho_2}(D_*)\setminus D_*)$, because $(1.1)^{2n}(1+\hat E_0/2\pi)^n\le(1.21\cdot1.01)^n<2^n$.

Step S4.34: The shell. $D_*\subset N_\eta(D_0)$, so $N_{\rho_2}(D_*)\subset N_{\rho_2+\eta}(D_0)\subset N_r(D)$ ($\rho_2+\eta\le 22\hat E_0\hat\rho_0<r$), and $N_{\rho_2}(D_*)\setminus D_*\subset(N_r(D)\setminus D)\cup(D_0\setminus D_*)\cup(D\setminus D_0)$: if $x\in N_{\rho_2}(D_*)\setminus D_*$ then either $x\notin D$ (first set), or $x\in D\setminus D_0$, or $x\in D_0\setminus D_*$. Its measure is $\le b+a+(4/3)^n(a+b)=(1+(4/3)^n)(a+b)$ by Lemma 4.9.

Step S4.35: Sum. Total $\le[a+(4/3)^n(a+b)]+2^n(1+(4/3)^n)(a+b)\le[1+(4/3)^n+2^n(1+(4/3)^n)](a+b)$. □

4.5. Proof of Proposition 3.3.

Proof of Proposition 3.3. **Step S4.36: Construction.** The sequence of Lemma 3.12 exists (hypotheses (S1)–(S3) at $j=0$ from the proposition's hypotheses and the definition of D_0); Lemma 4.2 defines $\check\imath$, Lemma 4.4 defines Φ, and φ_ω is defined in §4.3.

Step S4.37: (a), (d). Lemmas 4.4, 4.5.

Step S4.38: (b), (c). Lemmas 4.7, 4.6.

Step S4.39: Injectivity and Lipschitz structure. If $\varphi_\omega(\vartheta)-\varphi_{\omega'}(\vartheta')=(0,k)$, then $\Phi(I_*,\vartheta)=\Phi(I'_*,\vartheta'+k)$ by periodicity (Lemma 4.4(iii)), so $(I_*,\vartheta)=(I'_*,\vartheta'+k)$ by (ii); $I_*=I'_*$ gives $I_0=I'_0$ ($\check\imath$ injective), so $\omega=\omega_0(I_0)=\omega'$. The factorization and the constants: Lemma 4.2(iii) and Lemma 4.4(ii). For ω_0^{-1}: if $\omega,\omega'\in\omega_0(D)$ with $|\omega-\omega'|<\rho/\bar M$, then Lemma 3.6 (with $F=\omega_0$ on D_ρ, $\delta=\rho$, $x_0=\omega_0^{-1}(\omega)$) gives $|\omega_0^{-1}\omega-\omega_0^{-1}\omega'|\le\bar M|\omega-\omega'|$; for far pairs $|\omega_0^{-1}\omega-\omega_0^{-1}\omega'|\le\mathrm{diam}\,D\le\mathrm{diam}\,D\cdot\frac{\bar M}{\rho}|\omega-\omega'|$.

Step S4.40: Measure. $K_\Omega\subset\mathrm{Re}(D_r)\times\mathbb{T}^n$ (Lemma 4.5) so $(\mathrm{Re}(D_r)\times\mathbb{T}^n)\setminus K_\Omega=((\mathrm{Re}(D_r)\setminus D)\times\mathbb{T}^n)\cup((D\times\mathbb{T}^n)\setminus K_\Omega)$, and Lemma 4.10 gives (3.1) with $c_n=1+[1+(4/3)^n+2^n(1+(4/3)^n)]=1+(1+2^n)(1+(4/3)^n)$ ($a=\mathrm{meas}(D\setminus D^{\mathrm{dio}})$, $b=\mathrm{meas}(\mathrm{Re}(D_r)\setminus D)$, $\mathrm{meas}\,\mathbb{T}^n=1$). □

4.6. **What Part 5 consumes.** Part 5 ([P:bpP:p5:kam-application]) applies Proposition 3.3 to $H = \breve{H}$ of CP10 Step 6 (p. 560) with $n_1 = a$, $n_2 = b$, D the set $D = \omega_\mu^{-1}(\cdot) \cap \dots$ of CP10 (48), $h = H_0(J_1) + \mu\breve{N}(J)$, $f = \mu^5\breve{P}$, $\rho = \breve{\rho}$, $s = \breve{s}/5$, $\bar{s} = 4\breve{s}/5$. The data in the order the frozen quantifiers require:

(1) *Before* r, μ (depend only on the fixed family): n, τ (hence $\hat{c}, c_n$), the constants C with $M \le C$, $\hat{M} \le C\mu$, $\bar{M} \le C\mu^{-1}$, $\bar{M}_1 \le C$, $\bar{M}_2 \le C\mu^{-1}$ (CP10 p. 560, "$M = C, \hat{M} = C\mu, \bar{M} = C\mu^{-1}, E = C\mu^5, \bar{M}_1 = C, \bar{M}_2 = C\mu^{-1}$").

(2) *Checked for each* (r, μ) *with* $\mu < r^6/(\log(1/r))^c$: $\hat{c}\hat{E} < 1$ with $\hat{E} = EL/\hat{\rho}^2$, $K = \frac{6}{s}\log_+(\gamma_1^2/(EM^2L)) \le C\log(\epsilon^5/\gamma_1^2)^{-1}$; the normalizations $\hat{M} \le M$, $\bar{M}_i \le \bar{M}$ hold for μ small.

(3) *Outputs used*: the family φ_ω, $\omega \in \Omega_*$, with (a)–(d); the measure inequality (3.1) with c_n (an *absolute* constant, so the constant in meas$(\dots) \le C(\dots)$ of CP10 (55) is c_n); the radius $r = 32\hat{E}\hat{\rho}$ (CP10: $20n\hat{E}\hat{\rho}$; the constant changes nothing since $r \le C\hat{\rho}\hat{E} \le C\bar{\gamma}_2$); $|v - I^0|$ and $|u|$ for CP10 (51)-based estimates.

(4) *Conversions* (Part 5, [P:bpP:p1:period-convert]): frozen torus parameter $\theta \in \mathbb{R}^{a+b}$ of period one, fast angles of period 2π, frozen frequency $w = \omega/2\pi$ when CP's H is converted by Lemma 3.5; frozen order (position, momentum) = (angle, action).

Source gap: The quantifier *"for $\omega \in \Omega_*$, φ_ω unique"* of CP10 Prop. 3 is not used and is not claimed: translations $\vartheta \mapsto \vartheta + c$ preserve the conjugacy. CP10 Prop. 3 asserts that $(\vartheta, \omega) \mapsto \varphi_\omega(\vartheta)$ is *Lipschitz*; the argument only gives it through ω_0^{-1}, which is Lipschitz globally only for bounded D (locally, at scale $\rho/\bar{M}$, in general); see Proposition 3.3.

4.7. **Source gaps and suspected misprints (CP10, Prop. 3, Appendix B).**

(E1) p. 554: "$\hat{U}$ denote the $n \times n_1$ submatrix of U, i.e. the matrix with entries $\hat{U}_{ij} = U_{ij}$, for $n_1 + 1 \le i \le n$, $1 \le j \le n$": size and index range disagree; the proof needs the $n_2 \times n$ block (Remark 3.1).

(E2) p. 565 (Lemma B.1): "$M_0 := M$, $\hat{M}_0 := M$, $\bar{M}_0 := \bar{M}$": $\hat{M}_0$ should be $\hat{M}$.

(E3) p. 566, B.1(i): "$\hat{\imath}_j : I \in D_{j-1} \mapsto \omega_{j-1} \circ \omega_{j-1}^{-1}(I) \in D_j$" is the identity; it should read $\omega_j^{-1} \circ \omega_{j-1}$ (as in the proof of B.2, "$\hat{\imath}_+ := \omega_+^{-1} \circ \omega|_D$").

(E4) pp. 554–555, 566–568 (74), (77), (35): the inequalities $\bar{M}_i/\bar{M} \le 1$ (74) and $\hat{M}/M \le 1$ are used but not assumed (Remark 3.4). (35), second line: factor $\hat{M}/M$ is not derived (Lemma 4.5).

(E5) pp. 551, 566–570: the norms are mixed: "in $\mathbb{R}^{n_1}$ we fix the 1-norm, in $\mathbb{R}^{n_2}$ the sup norm, for matrices the sup-norm $|\beta| = \max|\beta_{ij}|$" (p. 551) but the proof uses operator norms $\|U\|, \|U^{-1}\|$, Cauchy bounds "$\sup\|\partial^2 g\| \le 2^7 E/\hat{\rho}^2$" (93) with no n and 1-/∞-norm conversions (e.g. $5n\bar{M}E/\hat{\rho}$ in (86)). The constants of (74), (75), (86), (87), (98), (104), (108) cannot all hold as printed; recomputed here in max norms.

(E6) p. 569 (91): "$\hat{E}_+ \le \frac{8}{9}(4s)^{\tau+1}\hat{E}^2$": the computation gives $\frac{8}{9}(4s)^{2(\tau+1)}\hat{E}^2$ (weaker printed bound is still true).

(E7) p. 569, injectivity of ω_+ (and (86)): "$\sup\|\partial(\omega^{-1})\| \sup|\omega(I_+) - \omega(I'_+)|$" needs the segment $[\omega(I), \omega(I')]$ inside $\omega(D_\rho)$ (not known; image sets are not convex): repaired by Lemma 3.6. The Lipschitz bounds on the non-convex sets D, D_j via "$\sup\|D(\hat{\imath} - \mathrm{id})\|$" are repaired by Lemma 3.7; the composition $\breve{\Psi}_1 \circ \dots \circ \breve{\Psi}_j$ (p. 572) needs $\Psi_k(\text{real domain}_k) \subset \text{real domain}_{k-1}$ (Lemma 4.3(i)).

(E8) p. 572–573 Step 3: boundedness of $D\Phi_j$ is not stated; supplied by the rescaled Lipschitz bounds (Step S4.22). Analyticity of φ_ω in the strip is not argued (Step S4.15, Vitali).

(E9) p. 555 and p. 573: "a unique real-analytic embedding" (not proved, not true as stated); Lipschitz continuity in ω (only locally/for bounded D).

(E10) p. 573 Step 4: Kirszbraun (Euclidean) is used for maps whose Lipschitz constants were computed in $1/\infty$ norms; footnote 33's "meas $f(A) \le L(f)^n$ meas A" is Euclidean. In max norm (McShane, cube covering) the factor is exactly Λ^d and no dimensional constant appears ([K:bp:p1:lip-image], [K:bp:p1:mcshane]); c_n is recomputed. The printed $c_n = (1 + (1 + 2^8 n\hat{E})^{2n})^2$ is not reproduced.

(E11) Prop. 1/Lemma A.1 (used in the proof of B.1 with $m = 0$, $\ell_1 = n_1$, $\ell_2 = n_2$): proved only by reference to [4, Lemma A.5] (unpinned), and the two-scale case is not covered by the plan of Part 2 (

Source gap:

at Lemma 3.8). Remark 3 (p. 567) "generalizes the inductive theorem of [3, p. 144]" is the only source for the scheme.

4.8. **Forward references and imported facts.** Forward references to *unwritten* parts: [N:bpN:p4:cauchy], [N:bpN:p4:bracket], [N:bpN:p4:projections], [N:bpN:p5:flow-domain], [N:bpN:p5:composition], [N:bpN:p5:taylor], [N:bpN:p5:symplectic], [N:bpN:p6:homological], [N:bpN:p6:iterative] (in the proof plan of Lemma 3.8 only); [P:bpP:p1:period-convert], [P:bpP:p2:averaging], [P:bpP:p5:kam-application]. Track K labels used: [K:bp:p1:cauchy-open], [K:bp:p1:cohomological], [K:bp:p1:uniform-limits], [K:bp:p1:mcshane], [K:bp:p1:lip-image], [K:bp:p1:ext-mvt], [K:bp:p1:ext-banach]. The averaging step Lemma 3.8 is the *only* unproved ingredient of this part; everything else is proved from the external facts (X1)–(X6).